prd

Pielot PRD/

1. Product Name

Pielot

2. Tagline

AI SMS revenue agent for local restaurants.

3. One-Liner

Pielot helps local restaurants increase repeat orders by analyzing customer behavior, finding revenue opportunities, and generating personalized SMS campaigns that owners can approve before sending.

4. Product Summary

Local restaurants, especially pizza shops, sit on valuable customer data: order history, favorite items, visit frequency, average order value, coupon usage, location, time-of-day behavior, and lapsed-customer patterns. But most owners do not have time to analyze this data, create customer segments, write marketing campaigns, test offers, or follow up with customers manually.

Pielot acts as an AI revenue pilot for the restaurant. It connects to ordering data, customer lists, POS exports, spreadsheets, and SMS tools. It identifies the highest-value revenue opportunities, such as lapsed customers, game-night buyers, family bundle buyers, discount-sensitive customers, high-value regulars, lunch customers, and abandoned cart users. Then it recommends the best offer, writes the SMS, predicts likely revenue impact, and prepares the campaign for owner approval.

Pielot does not blindly spam customers. It is built with consent, compliance, opt-out handling, frequency caps, human approval, and campaign audit logs as core product features.

5. Core Concept

Pielot is not just an SMS tool.

It is an **agentic restaurant revenue system**.

The product answers three questions:

1. **Who should we message?**
2. **What should we offer them?**
3. **When should we send it to maximize orders?**

Then it carries out the next step:

1. Creates customer segments.
2. Generates personalized offers.
3. Writes compliant SMS copy.
4. Estimates campaign impact.
5. Asks the owner for approval.
6. Sends or schedules the campaign.
7. Tracks orders, redemptions, opt-outs, and revenue lift.

6. Problem

Small restaurants lose revenue because they do not have a marketing team.

They often have:

- Past customers who have not ordered in weeks
- Regulars who could be upsold
- Families who would buy bundles
- Students who respond to late-night discounts
- Sports fans who order during games
- Coupon-sensitive customers who need the right incentive
- Happy customers who could be asked for referrals or reviews
- Customers who abandoned an order
- Customers who ordered once but never came back

But owners are busy handling staff, inventory, operations, food quality, and daily sales. Even when they have customer data, it usually lives across disconnected systems: Toast, Square, Clover, DoorDash exports, Uber Eats data, spreadsheets, Google Sheets, loyalty systems, email lists, and phone numbers.

The result: restaurants send generic blasts or no marketing at all.

Generic blasts are weak. No marketing leaves money on the table.

7. Target Users

Primary User: Local Restaurant Owner

Examples:

- Pizza shop owner
- Indian restaurant owner
- Taqueria owner
- Burger shop owner
- Cafe owner
- Food truck owner
- Small chain operator

Needs:

- More repeat customers
- More orders during slow hours
- Better customer retention
- Simple marketing without hiring a marketer
- Easy approval workflow
- Clear ROI

Pain points:

- Does not have time to analyze data
- Does not know what offer to send
- Does not know which customers to target
- Afraid of annoying customers
- Existing SMS tools feel too manual
- Agencies are expensive

Secondary User: Restaurant Manager

Needs:

- Fill empty time slots
- Promote specials
- Recover slow days
- Push inventory before waste
- Increase lunch/dinner rush

Secondary User: Marketing Consultant / Agency

Needs:

- Manage campaigns for multiple restaurants
- Use AI to generate segments and copy
- Prove campaign ROI

- Reduce manual reporting work

8. Ideal Customer Profile

Beachhead Customer

Independent pizza shops and small restaurant groups with:

- 500+ customer phone numbers
- Some order history
- Repeat purchase behavior
- Existing POS/export data
- Interest in SMS marketing
- No in-house marketing team

Why Pizza Shops First

Pizza shops are ideal because:

- Pizza has high repeat purchase behavior
- Offers are easy to understand
- Customer preferences are predictable
- Timing matters: Friday nights, sports games, lunch, late night
- Bundles are natural: pizza + wings + soda + garlic knots
- SMS works well for urgent food offers
- Demo is fun and memorable

9. Product Goals

MVP Goals

1. Upload or connect customer/order data.
2. Analyze customers and identify revenue opportunities.
3. Generate targeted SMS campaigns.
4. Show estimated revenue impact.
5. Require owner approval before sending.
6. Track campaign performance.
7. Handle opt-outs and consent status.

Hackathon Goals

1. Make the product instantly understandable.
2. Demo a fake pizza shop called “Pleasure Pizza.”

3. Show AI-generated customer segments.
4. Show personalized campaign recommendations.
5. Show an approval flow.
6. Show campaign results and projected revenue.
7. Emphasize safe, consent-based messaging.

Long-Term Goals

1. Become the default AI growth agent for local restaurants.
2. Integrate with POS systems.
3. Automate lifecycle campaigns.
4. Build restaurant-specific revenue intelligence.
5. Expand from SMS to email, WhatsApp, voice, and loyalty programs.
6. Provide multi-location analytics for small restaurant chains.

10. Non-Goals

For MVP, Pielot will not:

- Replace the restaurant's POS system
- Build a full CRM from scratch
- Handle payments directly
- Guarantee campaign revenue
- Send messages without consent
- Send messages without approval
- Support every restaurant category equally
- Build a complex loyalty program
- Build a full delivery marketplace
- Scrape private customer data
- Use sensitive personal attributes for targeting

11. Key Differentiator

Most SMS tools ask the owner:

"What message do you want to send?"

Pielot tells the owner:

"Here are the highest-value campaigns you should send today, who should receive them, why they are likely to work, and the exact message to approve."

The difference is that Pielot is proactive and agentic.

It does not just send campaigns. It finds revenue opportunities.

12. Core User Story

As a pizza shop owner, I want Pielot to look at my customer and order data, find the best revenue opportunities for today, generate targeted SMS campaigns, and let me approve them, so I can increase orders without manually doing marketing work.

13. Core Product Loop

1. Restaurant connects or uploads customer/order data.
2. Pielot analyzes buying behavior.
3. Pielot detects revenue opportunities.
4. Pielot generates campaign recommendations.
5. Owner approves, edits, or rejects campaigns.
6. Campaign is sent or scheduled.
7. Orders and redemptions are tracked.
8. Pielot learns which offers work best.
9. Better campaigns are recommended next time.

14. MVP Feature Set

Feature 1: Restaurant Setup

The owner creates a restaurant profile.

Fields:

- Restaurant name
- Cuisine type
- Location
- Hours
- Popular items
- Average order value
- Slow days
- Slow hours
- Current offers
- Brand tone
- SMS sender name
- Website/order link
- Consent policy status

Example setup:

Restaurant: Pleasure Pizza
Cuisine: Pizza
Location: Downtown Santa Cruz
Average order value: $24
Slow hours: Monday to Wednesday, 2 PM to 5 PM
Popular items: Pepperoni pizza, wings, garlic knots, family combo
Brand tone: Friendly, fun, local, casual

Feature 2: Data Upload

The owner uploads a CSV or connects a data source.

Supported MVP sources:

- CSV upload
- Google Sheets
- Manual sample data
- Mock POS data for demo

Future sources:

- Toast
- Square
- Clover
- Shopify
- DoorDash Storefront
- Uber Eats
- Stripe
- Loyalty systems

Required data columns:

- customer_id
- first_name
- phone_number
- sms_opt_in_status
- last_order_date
- order_count
- total_spend
- average_order_value
- favorite_items
- last_items_ordered
- coupon_usage
- preferred_order_day
- preferred_order_time

- tags
- last_campaign_received
- last_campaign_response
- opt_out_status

Optional data columns:

- birthday
- location/zip code
- household size estimate
- delivery/pickup preference
- review history
- abandoned cart status
- loyalty points
- favorite category
- dietary preference if explicitly provided by customer

Feature 3: Customer Segmentation Agent

The segmentation agent groups customers into revenue-relevant segments.

Example segments:

1. **Lapsed Customers**
    - No order in 30, 45, or 60 days
2. **Friday Regulars**
    - Usually order Friday evenings
3. **Game Night Buyers**
    - Order during sports events or weekends
4. **Family Bundle Buyers**
    - Order large pizzas, multiple items, or high-ticket orders
5. **Coupon Lovers**
    - High response to discounts
6. **High-Value Customers**
    - High total spend or high frequency
7. **Lunch Crowd**
    - Orders during lunch hours
8. **Late-Night Customers**
    - Orders after 9 PM
9. **Vegetarian Customers**
    - Frequently order veggie items
10. **Abandoned Order Customers**
- Started but did not complete an order
11. **New Customers**
- Ordered once but not again

12. **Churn Risk Customers**

- Used to order frequently but recently stopped

Feature 4: Opportunity Detection Agent

This agent identifies specific revenue opportunities.

Opportunity examples:

Win-Back Opportunity

Signal:
Customer has not ordered in 45+ days.

Recommended action:
Send comeback offer.

Example:
“143 customers have not ordered in 45+ days. Offer free garlic knots with a large pizza.”

Upsell Opportunity

Signal:
Customer frequently buys one pizza but never adds sides.

Recommended action:
Offer discounted wings or garlic knots bundle.

Example:
“62 pepperoni regulars never add sides. Offer pepperoni + wings combo.”

Timing Opportunity

Signal:
Customer typically orders on Friday nights.

Recommended action:
Send offer Friday at 4:30 PM.

Example:
“87 customers usually order on Friday evenings. Send family bundle before dinner.”

Slow-Hour Opportunity

Signal:
Restaurant has weak sales during Monday lunch.

Recommended action:
Send lunch special to nearby/lunch customers.

Example:
“Lunch sales are slow on Mondays. Send $9 slice + drink offer to lunch customers.”

Event-Based Opportunity

Signal:
Sports game or local event.

Recommended action:
Send game-night bundle.

Example:
“Warriors game tonight. Send pizza + wings bundle to sports-night buyers.”

Review/Referral Opportunity

Signal:
Customer is happy or high frequency.

Recommended action:
Ask for review or referral.

Example:
“54 customers ordered 5+ times and never received referral message.”

Abandoned Cart Opportunity

Signal:
Customer started order but did not complete.

Recommended action:
Send reminder or small incentive.

Example:
“23 customers abandoned checkout this week. Send 10% off if they finish today.”

Feature 5: Offer Generation Agent

The offer agent chooses the best offer for each segment.

Offer types:

- Free garlic knots
- Free soda

- 10% off
- 20% off
- Buy one get one 50% off
- Family combo
- Lunch special
- Student late-night deal
- Game-night bundle
- Free delivery
- Loyalty points booster
- Birthday offer
- Comeback coupon
- Referral reward
- First reorder discount

Offer selection factors:

- Customer price sensitivity
- Average order value
- Past coupon usage
- Last order date
- Favorite items
- Day/time behavior
- Restaurant margin settings
- Inventory availability
- Campaign goal

Offer constraints:

- Do not offer discounts below margin threshold.
- Do not send multiple offers to the same customer too frequently.
- Do not send irrelevant offers.
- Do not use sensitive attributes.
- Do not target opted-out customers.
- Do not generate misleading urgency.

Feature 6: SMS Copy Agent

The SMS agent writes short, compliant, brand-matched messages.

Inputs:

- Customer segment
- Offer
- Restaurant name
- Tone

- Order link
- Opt-out text
- Campaign goal

SMS requirements:

- Clear sender identity
- Clear offer
- Clear call to action
- Short message
- No deceptive urgency
- Opt-out language
- Human approval before sending

Example messages:

Win-Back

Pleasure Pizza: We miss you. Get free garlic knots with any large pizza tonight. Order here: [link]. Reply STOP to opt out.

Game Night

Pleasure Pizza: Game night bundle is live. Large pepperoni + wings for $22 tonight only. Order here: [link]. Reply STOP to opt out.

Family Friday

Pleasure Pizza: Friday family deal: 2 large pizzas + soda for $29. Perfect for dinner tonight. Order here: [link]. Reply STOP to opt out.

Late Night

Pleasure Pizza: Late-night slice deal is on. Grab 2 slices + drink for $7 after 9 PM. Order here: [link]. Reply STOP to opt out.

Feature 7: Campaign Approval Flow

Before sending, Pielot presents:

- Campaign name
- Target segment
- Number of recipients
- Why this segment was chosen
- Offer
- SMS copy

- Estimated revenue
- Estimated cost
- Estimated margin
- Send time
- Compliance checklist
- Frequency cap warning
- Opt-out count risk
- Approval button

Owner actions:

- Approve campaign
- Edit SMS
- Change offer
- Change audience
- Schedule later
- Reject campaign

Feature 8: Campaign Execution

MVP execution options:

- Simulated sending for demo
- Real Twilio integration for approved opt-in test numbers
- Export campaign as CSV
- Copy SMS text
- Schedule send

Future execution:

- Twilio
- Klaviyo
- Attentive
- Postscript
- Square Marketing
- Toast Marketing
- Mailchimp SMS
- WhatsApp Business

Feature 9: Campaign Performance Tracking

Track:

- Sent
- Delivered

- Failed
- Clicks
- Replies
- Opt-outs
- Orders
- Revenue
- Average order value
- Redemption rate
- Revenue per message
- Gross margin estimate
- Campaign ROI

Dashboard metrics:

- Total campaign revenue
- Revenue lift
- Best performing segment
- Best performing offer
- Best send time
- Opt-out rate
- Repeat purchase rate
- Lapsed customers recovered

Feature 10: Agent Activity Log

Every agent action is logged.

Example log:

- Segmentation Agent analyzed 412 customers.
- Opportunity Agent found 6 campaign opportunities.
- Offer Agent selected free garlic knots for lapsed customers.
- SMS Agent generated 3 message variants.
- Compliance Agent confirmed opt-in-only audience.
- Owner approved campaign.
- Campaign scheduled for 4:30 PM.
- 143 messages sent.
- 21 orders attributed.
- $612 revenue generated.

15. Agent Architecture

Pielot is made of multiple specialized agents coordinated by a main orchestrator.

Main Agent: Revenue Pilot Agent

Responsibilities:

- Understand business context
- Coordinate sub-agents
- Rank opportunities
- Produce final campaign recommendations
- Explain reasoning to owner

Agent 1: Data Ingestion Agent

Responsibilities:

- Read uploaded CSV or connected source
- Validate columns
- Normalize customer/order data
- Detect missing fields
- Flag bad phone numbers
- Remove opted-out users
- Prepare clean dataset

Agent 2: Customer Segmentation Agent

Responsibilities:

- Create customer segments
- Score each customer
- Assign customers to campaign audiences
- Explain segment logic

Agent 3: Opportunity Detection Agent

Responsibilities:

- Find revenue opportunities
- Detect lapsed customers
- Detect upsell moments
- Detect slow-hour campaigns
- Detect event-based campaigns
- Detect abandoned carts
- Detect referral opportunities
- Estimate potential revenue

Agent 4: Offer Strategy Agent

Responsibilities:

- Choose offer type
- Estimate margin impact
- Avoid over-discounting
- Personalize offer by segment
- Recommend best send time

Agent 5: SMS Copy Agent

Responsibilities:

- Generate SMS text
- Match restaurant tone
- Create variants
- Include call to action
- Keep message concise
- Include opt-out language

Agent 6: Compliance Guardrail Agent

Responsibilities:

- Filter opted-out users
- Verify opt-in status
- Check frequency caps
- Check required opt-out text
- Prevent misleading claims
- Prevent sensitive targeting
- Require human approval

Agent 7: Campaign Execution Agent

Responsibilities:

- Schedule campaign
- Send via SMS provider
- Record send status
- Track replies
- Process STOP/opt-out
- Update campaign status

Agent 8: Learning Agent

Responsibilities:

- Analyze campaign performance
- Learn which offers worked
- Recommend future campaigns
- Update customer profiles
- Suggest next best action

16. System Architecture

MVP Architecture

Frontend:

- Next.js or React
- Tailwind CSS
- Dashboard UI
- Campaign approval modals

Backend:

- Node.js/Express or FastAPI
- OpenAI Agents SDK or OpenAI API
- Campaign recommendation endpoint
- Mock Twilio send endpoint
- CSV parser

Database:

- Supabase/Postgres
- Tables for restaurants, customers, orders, campaigns, messages, opt-outs, agent logs

Integrations:

- CSV upload
- Google Sheets optional
- Twilio optional
- Mock POS data

Production Architecture

Frontend:

- Next.js
- Tailwind
- Shadcn UI
- Recharts for analytics

Backend:

- FastAPI or Node.js
- Agent orchestration service
- Queue for campaign jobs
- SMS provider adapter
- POS integration service

Database:

- Postgres
- Redis for queues/cache
- Vector database optional for business memory

Storage:

- S3/Supabase Storage for CSVs and exports

Auth:

- Supabase Auth
- Restaurant workspace accounts
- Owner/admin roles

Monitoring:

- Campaign delivery logs
- Agent audit logs
- Error tracking
- SMS provider webhook logs

17. Data Model

restaurants

Fields:

- id
- name
- cuisine_type
- location
- timezone
- average_order_value
- brand_tone
- order_link

- sms_sender_name
- created_at
- updated_at

customers

Fields:

- id
- restaurant_id
- first_name
- phone_number
- sms_opt_in_status
- opt_out_status
- last_order_date
- order_count
- total_spend
- average_order_value
- favorite_items
- preferred_order_day
- preferred_order_time
- coupon_usage_score
- customer_lifecycle_stage
- created_at
- updated_at

orders

Fields:

- id
- restaurant_id
- customer_id
- order_date
- order_time
- items
- subtotal
- discount_used
- channel
- delivery_or_pickup
- created_at

segments

Fields:

- id
- restaurant_id
- name
- description
- criteria_json
- customer_count
- created_at

campaigns

Fields:

- id
- restaurant_id
- name
- segment_id
- campaign_type
- offer
- sms_body
- status
- estimated_revenue
- estimated_margin
- projected_orders
- scheduled_at
- approved_by
- approved_at
- created_at

campaign_recipients

Fields:

- id
- campaign_id
- customer_id
- phone_number
- delivery_status
- clicked
- replied
- ordered
- revenue_attributed
- opted_out_after
- sent_at

agent_logs

Fields:

- id
- restaurant_id
- agent_name
- action
- input_summary
- output_summary
- created_at

opt_outs

Fields:

- id
- restaurant_id
- phone_number
- opt_out_keyword
- opt_out_at
- source_campaign_id

18. Scoring Logic

Pielot should score opportunities using a simple model first.

Customer Buy Likelihood Score

Inputs:

- Recency of last order
- Frequency of orders
- Average order value
- Preferred time match
- Past coupon response
- Favorite item match
- Campaign fatigue
- Segment relevance

Example formula:

Buy Likelihood =
0.25 × Recency Score

- 0.20 × Frequency Score
- 0.15 × Offer Match Score
- 0.15 × Preferred Timing Score
- 0.10 × Coupon Response Score
- 0.10 × Favorite Item Match Score
- 0.15 × Fatigue Penalty

Opportunity Priority Score

Inputs:

- Potential revenue
- Number of customers
- Urgency
- Margin impact
- Likelihood to buy
- Strategic value
- Risk of opt-out

Example:

Priority Score =
Potential Revenue × Buy Likelihood × Margin Score × Urgency Multiplier - Fatigue Risk

19. Example Campaign Recommendations

Campaign 1: Win Back Lapsed Customers

Audience:
Customers who have not ordered in 45+ days.

Customer count:
143

Offer:
Free garlic knots with any large pizza.

Why:
These customers previously ordered but have gone inactive. A low-cost side offer may reactivate them without heavily discounting the main item.

SMS:
Pleasure Pizza: We miss you. Get free garlic knots with any large pizza tonight. Order here: [link]. Reply STOP to opt out.

Projected results:

- Expected orders: 28
- Average order value: $22
- Projected revenue: $616
- Estimated food cost: $170
- Estimated gross profit: $446

Campaign 2: Game Night Bundle

Audience:
Customers who order on weekends or during previous sports nights.

Customer count:
87

Offer:
Large pepperoni pizza + wings for $22.

Why:
These customers have high intent during game-night windows and respond well to bundles.

SMS:
Pleasure Pizza: Game night bundle is live. Large pepperoni + wings for $22 tonight. Order here: [link]. Reply STOP to opt out.

Projected results:

- Expected orders: 31
- Average order value: $22
- Projected revenue: $682
- Estimated gross profit: $420

Campaign 3: Family Friday

Audience:
Customers who buy multiple pizzas or large orders.

Customer count:
64

Offer:
2 large pizzas + soda for $29.

Why:
These customers already purchase family-sized meals. A bundle may increase conversion and order size.

SMS:
Pleasure Pizza: Friday family deal: 2 large pizzas + soda for $29. Dinner solved. Order here: [link]. Reply STOP to opt out.

Projected results:

- Expected orders: 18
- Average order value: $29
- Projected revenue: $522

Campaign 4: Late Night Student Deal

Audience:
Customers who order after 9 PM.

Customer count:
118

Offer:
2 slices + drink for $7 after 9 PM.

Why:
Late-night buyers are timing-sensitive and respond to simple, immediate offers.

SMS:
Pleasure Pizza: Late-night deal after 9 PM: 2 slices + drink for $7. Order here: [link]. Reply STOP to opt out.

Projected results:

- Expected orders: 35
- Average order value: $11
- Projected revenue: $385

20. User Flow

Flow 1: First-Time Setup

1. Owner signs up.
2. Owner creates restaurant profile.
3. Owner uploads customer/order CSV.

4. Pielot validates data.
5. Pielot shows data quality report.
6. Owner confirms consent/opt-in source.
7. Pielot generates first revenue opportunities.
8. Owner reviews recommended campaigns.
9. Owner approves one campaign.
10. Campaign is scheduled or sent.

Flow 2: Daily Revenue Scan

1. Owner opens dashboard.
2. Pielot automatically scans latest data.
3. Pielot shows “Today’s Revenue Opportunities.”
4. Owner reviews campaigns.
5. Owner approves best campaign.
6. Pielot sends campaign.
7. Dashboard updates with performance.

Flow 3: Campaign Approval

1. Owner clicks campaign card.
2. Pielot explains why campaign was recommended.
3. Owner reviews audience, offer, message, and send time.
4. Owner edits message if needed.
5. Compliance checklist runs.
6. Owner clicks Approve.
7. Campaign is scheduled.
8. Campaign status is tracked.

Flow 4: Opt-Out Handling

1. Customer replies STOP or similar opt-out keyword.
2. Pielot marks customer as opted out.
3. Customer is excluded from future campaigns.
4. Owner can see opt-out count in campaign report.
5. Agent adjusts future campaign frequency if opt-out rate is high.

21. UI Screens

Screen 1: Landing Page

Hero:

Pielot
AI SMS revenue agent for local restaurants.

Subtext:

Pielot analyzes customer behavior, finds revenue opportunities, and creates personalized SMS campaigns that bring customers back.

CTA:

- Launch Revenue Agent
- View Demo

Hero visual:

Dashboard showing:

- Revenue opportunities found
- Campaigns ready for approval
- Projected revenue

Screen 2: Restaurant Setup

Sections:

- Restaurant details
- Brand tone
- Popular items
- Slow hours
- Campaign goals
- Upload customer/order data

CTA:

- Analyze My Customers

Screen 3: Data Upload

Components:

- CSV upload box
- Google Sheets connect button
- Sample data button
- Data validation table
- Consent status checker

States:

- Uploading
- Validating
- Missing columns warning
- Opt-in status warning
- Data ready

Screen 4: Agent Scan

Visual progress:

- Reading order history
- Finding customer segments
- Detecting lapsed customers
- Matching offers to buying behavior
- Estimating campaign revenue
- Checking compliance
- Preparing campaigns

CTA after complete:

- View Opportunities

Screen 5: Main Dashboard

Top cards:

- Projected revenue
- Customers analyzed
- Campaigns ready
- Opted-in customers
- High-value opportunities

Main sections:

1. Revenue opportunities
2. Customer segments
3. Campaign recommendations
4. Agent activity log
5. Recent campaign performance

Screen 6: Campaign Detail

Information shown:

- Campaign name
- Audience
- Customer count
- Why this campaign
- Offer
- SMS copy
- Send time
- Projected revenue
- Compliance checklist
- Approval controls

Buttons:

- Approve Campaign
- Edit Message
- Change Offer
- Schedule Later
- Reject

Screen 7: Campaign Performance

Charts and metrics:

- Sent
- Delivered
- Clicked
- Replied
- Ordered
- Revenue
- Opt-outs
- ROI
- Best customer segment
- Best offer

Screen 8: Customer Segments

Cards:

- Lapsed customers
- Friday regulars
- Game-night buyers
- Family buyers
- Coupon lovers
- High-value customers
- New customers

- Late-night customers

Each card shows:

- Count
- Average order value
- Last campaign
- Recommended offer
- Revenue potential

Screen 9: Compliance Center

Sections:

- Opt-in status
- Opt-out list
- Frequency caps
- Message templates
- Campaign approval logs
- SMS provider status

22. Compliance and Safety Requirements

Pielot must be compliance-first.

Required Guardrails

1. Only message opted-in customers.
2. Never message opted-out customers.
3. Include opt-out language in campaign texts.
4. Respect STOP and similar opt-out replies.
5. Enforce frequency caps.
6. Require owner approval before sending.
7. Maintain audit logs.
8. Avoid sensitive-attribute targeting.
9. Avoid deceptive urgency.
10. Avoid misleading offers.
11. Show estimated cost and margin impact.
12. Allow owner to pause all campaigns.

Frequency Caps

Default:

- Max 2 marketing texts per customer per week
- Max 6 marketing texts per customer per month
- No messages before 9 AM or after 8 PM local time by default
- No repeat campaigns to same segment within 72 hours

Sensitive Targeting Restrictions

Do not target based on:

- Race
- Religion
- Health status
- Political affiliation
- Immigration status
- Precise sensitive location
- Financial hardship
- Children/minors
- Any sensitive protected attribute

Allowed targeting:

- Order history
- Favorite item
- Last order date
- Coupon usage
- Order frequency
- Average order value
- Preferred order day/time
- Explicitly provided preferences
- Loyalty status

23. AI Requirements

Model Behavior

The AI must:

- Explain why each campaign is recommended.
- Produce structured outputs.
- Avoid hallucinating customer data.
- Only use provided data.
- Ask for missing required fields.
- Use conservative revenue estimates.
- Respect compliance rules.

- Recommend no campaign if data quality is poor.
- Avoid manipulative copy.
- Generate brand-safe SMS messages.

Structured Campaign Output

The agent should return:

- campaign_name
- segment_name
- customer_count
- segment_reason
- offer
- sms_body
- send_time
- projected_orders
- projected_revenue
- estimated_margin
- risk_level
- compliance_status
- required_owner_action

24. Example Agent Output

Campaign Name:
Win Back Lapsed Pizza Customers

Segment:
Customers who have not ordered in 45+ days

Customer Count:
143

Why:
These customers previously purchased but have not returned recently. A low-cost side offer may bring them back without discounting the main product.

Offer:
Free garlic knots with any large pizza

SMS:
Pleasure Pizza: We miss you. Get free garlic knots with any large pizza tonight. Order here: [link]. Reply STOP to opt out.

Projected Orders:
28

Projected Revenue:
$616

Compliance Status:
Ready for approval. All recipients are opted in. Opt-out language included. Frequency cap passed.

Required Action:
Owner approval required before sending.

25. Hackathon MVP Build Plan

Build Name

Pielot Demo: Pleasure Pizza

Demo Restaurant

Pleasure Pizza, a local pizza shop.

Mock Data

Use 400–500 fake customers.

Customer examples:

- Ravi: orders pepperoni every Friday
- Maya: has not ordered in 60 days
- Alex: always uses coupons
- Priya: orders family meals
- Jordan: orders late-night slices
- Sam: orders during Warriors games
- Nina: abandoned checkout
- Chris: high-value regular
- Emily: vegetarian customer
- Omar: new customer who never reordered

MVP Screens to Build

1. Landing page
2. Data upload/mock connect screen

3. Agent scanning screen
4. Revenue opportunity dashboard
5. Campaign approval modal
6. Campaign performance screen

MVP Backend

Can be mocked.

Endpoints:

- POST /api/analyze
- GET /api/opportunities
- POST /api/generate-campaign
- POST /api/approve-campaign
- GET /api/campaign-results

MVP Agent Logic

Use OpenAI to:

- Analyze customer JSON
- Create segments
- Recommend campaigns
- Generate SMS copy
- Explain reasoning

Use deterministic fallback rules if needed:

- If last_order_date > 45 days, assign to lapsed segment.
- If preferred_order_day = Friday, assign to Friday regular.
- If coupon_usage_score > 0.7, assign to discount lover.
- If average_order_value > $35, assign to family/high-value.
- If preferred_order_time > 9 PM, assign to late-night.

26. Hackathon Demo Script

Opening:

Local restaurants do not need another dashboard. They need more orders. The problem is that their best revenue opportunities are hidden inside customer behavior: who has not ordered recently, who buys on Friday nights, who loves coupons, who orders during games, and who is likely to come back with the right offer.

What we built:

Pielot is an AI SMS revenue agent for local restaurants. It analyzes order history, finds customer segments, recommends personalized campaigns, writes the SMS, checks compliance, and asks the owner for approval.

Demo:

Here is Pleasure Pizza. We uploaded 412 customer records. Pielot scans the data and finds six revenue opportunities.

It found a win-back campaign for 143 lapsed customers, a game-night bundle for 87 customers, a family Friday deal for 64 customers, and a late-night student campaign for 118 customers.

Let’s open the win-back campaign. Pielot explains why this audience was selected, recommends free garlic knots with a large pizza, writes the message, estimates $616 in revenue, and confirms every recipient is opted in. The owner can edit, approve, or schedule the campaign.

Why this matters:

Most small restaurants cannot afford a marketing team. Pielot gives them an AI revenue pilot that finds opportunities and turns them into action.

Closing:

Pielot helps restaurants send the right offer to the right customer at the right time.

27. Success Metrics

Business Metrics

- Revenue generated per campaign
- Incremental order count
- Repeat purchase rate
- Customer reactivation rate
- Average order value lift
- Revenue per SMS
- Gross margin impact
- Campaign ROI

Product Metrics

- Campaigns approved per restaurant per week
- Time from data upload to campaign recommendation
- Owner edit rate
- Owner approval rate

- Segment accuracy
- Offer acceptance rate
- SMS click/reply rate
- Opt-out rate

Safety Metrics

- Opt-out rate by campaign
- Complaints
- Frequency cap violations
- Messages blocked by compliance agent
- Campaigns requiring manual review
- Invalid phone number rate

28. Pricing

Hackathon Pricing Slide

Starter:
$49/month
For small restaurants
Includes 1,000 SMS recommendations/month

Growth:
$149/month
For busy restaurants
Includes advanced segmentation, campaign analytics, and automated weekly campaigns

Multi-Location:
$399/month+
For restaurant groups
Includes multi-store analytics, team roles, and advanced integrations

Usage:
SMS sending cost billed separately or passed through.

Alternative Performance Pricing

Pielot takes a small percentage of revenue generated from campaigns.

Example:
$99/month + 3% of attributed campaign revenue.

29. Go-To-Market

Initial GTM

1. Walk into local pizza shops.
2. Offer free campaign audit.
3. Upload their past order CSV.
4. Show “found revenue opportunities.”
5. Let them approve one campaign.
6. Track orders.
7. Convert to monthly subscription.

Sales Pitch

“You already have customers who would order again if they got the right offer. Pielot finds them and writes the campaign for you.”

First Customer Acquisition Strategy

- Downtown restaurants
- College-town pizza shops
- Food trucks
- Local restaurant associations
- POS reseller partnerships
- SMS marketing agency partnerships
- Case study with one pizza shop

Case Study Goal

“Pielot generated $1,200 in additional orders for a local pizza shop in one weekend.”

30. Competitive Landscape

Existing Tools

- Twilio
- Klaviyo
- Postscript
- Attentive
- Mailchimp
- Toast Marketing
- Square Marketing
- Popmenu

- Owner.com
- Thanx

Pielot Difference

Existing tools help send campaigns.

Pielot finds what campaigns to send.

It is not just infrastructure. It is an AI revenue agent.

31. Risks

Risk 1: Compliance

SMS marketing can create legal risk.

Mitigation:
Opt-in only, opt-out handling, frequency caps, audit logs, human approval, compliance review.

Risk 2: Bad Data

Restaurants may have incomplete or messy data.

Mitigation:
CSV validation, missing-field warnings, conservative recommendations, sample templates.

Risk 3: Annoying Customers

Too many texts can hurt the restaurant brand.

Mitigation:
Frequency caps, fatigue scoring, opt-out monitoring, campaign quality scoring.

Risk 4: Weak Attribution

It may be hard to prove revenue came from SMS.

Mitigation:
Unique links, promo codes, redemption tracking, time-window attribution.

Risk 5: Over-Discounting

Bad offers can reduce margin.

Mitigation:
Margin-aware offer engine, owner-set discount limits, profit estimate.

32. Future Features

Near-Term

- Real Twilio integration
- Google Sheets integration
- CSV template generator
- Offer A/B testing
- Promo code generation
- Campaign calendar
- AI brand voice settings
- Multi-location support

Mid-Term

- POS integrations
- Toast/Square/Clover sync
- Loyalty program integration
- Inventory-aware campaigns
- Weather-based campaigns
- Sports/event-based campaigns
- Birthday campaigns
- Review request campaigns
- Referral campaigns
- Automated weekly revenue scan

Long-Term

- Fully autonomous lifecycle marketing
- Cross-channel campaigns: SMS, email, WhatsApp, push
- Voice AI campaign assistant
- Restaurant revenue forecasting
- Dynamic pricing suggestions
- AI-generated menu promotions
- Local event intelligence
- Multi-location benchmarking
- Agency dashboard

33. Example Future Automation

Every Friday at 2 PM, Pielot runs a revenue scan.

It detects:

- Friday dinner regulars
- Customers who have not ordered in 30 days
- Families with high average order value
- Sports-night customers
- Overloaded inventory items

Then it recommends:

1. Family Friday campaign
2. Game night bundle
3. Lapsed customer comeback offer
4. Inventory-clearance special

The owner opens the dashboard, approves two campaigns, and Pielot schedules them automatically.

34. MVP Acceptance Criteria

Pielot MVP is complete when:

1. A user can create a restaurant profile.
2. A user can upload or load mock customer data.
3. The system can segment customers.
4. The system can detect at least 4 revenue opportunities.
5. The system can generate SMS campaigns.
6. The system can show projected revenue.
7. The system can show compliance status.
8. The user can approve or reject a campaign.
9. The system can show an agent activity log.
10. The system can show mock campaign performance.

35. Hackathon Judging Narrative

Pielot is strong because it is:

- Specific: local restaurants
- Valuable: more revenue
- Demoable: generated SMS campaigns
- Agentic: detects, decides, drafts, and executes
- Safe: opt-in, approval, opt-out, audit logs

- Timely: small businesses want AI that makes money now

The judge takeaway should be:

“Pielot helps restaurants make more money from customers they already have.”

36. Final Product Statement

Pielot is the AI revenue pilot for local restaurants. It analyzes customer behavior, finds hidden revenue opportunities, creates personalized SMS campaigns, and lets owners approve campaigns in one click. Instead of making restaurant owners become marketers, Pielot gives them an AI agent that knows who to message, what to offer, and when to send it.

finance info

Customer Retention Optimization Plan

Using POS Data to Promote Purchases on Low-Demand Days

Executive Summary

This product is designed to help businesses improve **customer retention, repeat purchasing, and revenue consistency** by identifying low-demand days through POS data and promoting targeted offers to customers most likely to return.

The goal is not simply to discount products. The goal is to use POS behavior, customer purchase history, demand trends, and profitability signals to answer one key question:

> **Who should we encourage to buy, on which low-demand day, with what offer, and at what margin-safe discount?**

The platform should help businesses avoid slow-day revenue dips while protecting profit margin, customer trust, and brand value.

---

1. Product Objective

Core Goal

Create an optimized retention strategy that increases customer visits and purchases on low-demand days without over-discounting or attracting only deal-seeking customers.

Business Problem

Many businesses experience predictable low-demand periods, such as:

- Mondays or Tuesdays
- Midday slow periods
- Post-holiday dips
- Bad-weather days
- Off-season weeks

- Days without events, promotions, or high foot traffic

These slow periods create underused staff capacity, lower daily revenue, and inconsistent customer engagement.

Product Solution

Use POS data to detect low-demand windows, segment customers, recommend personalized promotions, and measure whether promotions actually improve retention and incremental revenue.

---

2. Strategic Model Overview

The optimal model should be a **promotion optimization and retention intelligence model**.

POS Transaction Data

↓

Demand Pattern Detection

↓

Customer Segmentation

↓

Promotion Recommendation Engine

↓

Low-Demand Day Campaigns

↓

Retention, Revenue, and Margin Measurement

What the Model Should Decide

Decision	Purpose

Which days are low-demand?	Identify when the business needs traffic
Which customers should be targeted?	Avoid wasting offers on customers who would buy anyway
What offer should be used?	Match the incentive to customer behavior
How large should the discount be?	Protect the margin while creating action
When should the promotion be sent?	Maximize conversion likelihood
Did the campaign actually work?	Measure true incremental impact

---

3. POS Data Needed

The system can work without manually inspecting raw customer data. It should use structured POS events and aggregated customer behavior.

Core POS Inputs

Data Type	Examples	Why It Matters
Transaction History	Orders, timestamps, basket size	Identifies purchase behavior
Customer ID	Loyalty ID, phone/email hash, POS customer profile	Connects repeat purchases

Product/SKU Data	Items purchased, categories, modifiers	Personalizes offers
Revenue Data	Gross sales, net sales, refunds, discounts	Measures financial impact
Visit Frequency	Days between purchases, repeat visits	Predicts retention risk
Day/Time Data	Weekday, hour, season, holiday	Detects low-demand periods
Location Data	Store, region, terminal	Supports store-level campaigns
Promotion History	Prior discounts, redemptions	Prevents over-targeting
Margin Data	Product cost, gross margin	Keeps offers profitable

Privacy-Preserving Approach

The model should not need to display sensitive customer-level records. It can operate on:

- Masked customer IDs
- Customer segments
- Aggregated behavior
- Purchase frequency bands
- Average order value ranges
- Product category preferences
- Retention-risk scores
- Promotion-response scores

---

4. Low-Demand Day Detection

The first model layer identifies when demand is weaker than expected.

What Counts as a Low-Demand Day?

A low-demand day is not just a day with low sales. It is a day where sales are below what the business should reasonably expect based on normal patterns.

The model should compare each day against:

- Same weekday historical average
- Same season or month
- Local holidays or events
- Weather conditions if relevant
- Prior promotion activity
- Store/location patterns
- Product category trends
- Customer traffic patterns

Example Output

Low-Demand Opportunity

Day: Tuesday

Time Window: 2 PM – 6 PM

Expected Revenue: $4,800

Predicted Revenue Without Promotion: $3,650

Demand Gap: -24%

Recommended Campaign: Repeat-customer afternoon offer

Estimated Incremental Revenue: $740

Margin Risk: Low

---

5. Customer Segmentation Strategy

The product should not send the same discount to every customer. Customers should be segmented based on behavior and likelihood to respond.

Recommended Segments

Segment	Definition	Best Promotion Type
Loyal Customers	Frequent repeat buyers	Small appreciation reward
At-Risk Customers	Used to buy often but have slowed down	Win-back offer
New Customers	Purchased once recently	Second-purchase incentive
High-AOV Customers	Spend more than average	Premium bundle or exclusive offer
Discount-Sensitive Customers	Usually buy with promotions	Controlled limited-time offer
Category Loyalists	Repeatedly buy same category	Category-specific offer
Dormant Customers	No purchase for a long period	Stronger reactivation offer
Low-Margin Customers	Buy low-profit items or require heavy discounts	Avoid deep discounts

Key Retention Metrics

Metric	Purpose

Repeat Purchase Rate	Measures how many customers come back
Purchase Frequency	Shows how often customers buy
Days Since Last Purchase	Identifies at-risk customers
Customer Lifetime Value	Estimates long-term customer value
Average Order Value	Measures transaction quality
Redemption Rate	Shows campaign response
Incremental Revenue	Measures revenue that would not have happened otherwise
Gross Margin After Discount	Confirms profitability
Retention Lift	Measures improvement versus control group

---

6. Promotion Recommendation Model

The core model should recommend the right offer for the right customer on the right low-demand day.

Model Inputs

Customer purchase history

Visit frequency

Average order value

Favorite categories

Days since last purchase

Prior promotion response

Discount sensitivity

Preferred purchase day/time

Location behavior

Gross margin by product/category

Current low-demand window

Inventory or capacity availability

Model Outputs

Recommended customer segment

Recommended promotion type

Recommended discount level

Best send time

Expected redemption rate

Expected incremental revenue

Expected gross margin impact

Risk of cannibalization

Confidence score

Recommended Model Types

Use Case	Model Type
Predict low-demand days	Time-series forecasting model
Identify customers likely to return	Churn / retention prediction model
Choose best promotion	Uplift modeling or treatment-effect model
Protect margin	Profit optimization model
Detect over-discounting	Anomaly detection model
Explain recommendations	AI assistant over approved metrics

Why Uplift Modeling Matters

A normal model may tell you who is likely to buy. But that can waste promotions on people who would have bought anyway.

An uplift model tries to identify customers who are more likely to buy **because of the promotion**.

That is the important difference:

Bad Targeting:

Send discount to customers who were already going to buy.

Better Targeting:

Send promotion to customers whose behavior is likely to change because of the offer.

7. Promotion Types

Promotions should be matched to customer behavior and margin constraints.

Recommended Offer Library

Promotion Type	Best For	Example
Second-Purchase Offer	New customers	“Come back Tuesday and get 10% off your next visit.”
Win-Back Offer	At-risk or dormant customers	“We miss you — enjoy $5 off this Wednesday.”
Category Offer	Customers with clear preferences	“Your favorite bowl is 15% off Monday lunch.”
Bundle Offer	Margin protection	“Buy a meal combo and save $3.”
Loyalty Multiplier	Loyal customers	“Earn 2x points on slow-day visits.”
Time-Window Offer	Low-demand hours	“Visit between 2–5 PM for a weekday reward.”
Inventory-Aware Offer	Perishable or excess inventory	“Today only: featured item special.”
Referral Offer	Loyal customers	“Bring a friend on Tuesday and both get a reward.”

Margin-Safe Promotion Rule

Every offer should pass a profitability check before launch.

Expected Incremental Revenue

- Discount Cost

- Product Cost

- Payment Fees

- Campaign Cost

= Expected Incremental Profit

The platform should reject or flag promotions that drive revenue but destroy margin.

---

8. Campaign Workflow

The product should make the campaign process simple and measurable.

Step 1: Detect Opportunity

The system identifies low-demand days and time windows.

Tuesday afternoon demand is projected to be 24% below baseline.

Step 2: Select Customer Segment

The model recommends which customers to target.

Target: At-risk customers who previously purchased on weekdays and have not returned in 21–45 days.

Step 3: Recommend Offer

The model chooses a promotion that balances conversion and margin.

Offer: $5 off orders above $30 between 2 PM and 6 PM.

Step 4: Approve Campaign

The business reviews expected impact.

Expected Redemption Rate: 11.8%

Expected Incremental Revenue: $740

Expected Gross Margin After Discount: 58%

Cannibalization Risk: Medium

Confidence: 86%

Step 5: Launch Campaign

Campaign is sent through SMS, email, app notification, receipt QR code, or loyalty platform.

Step 6: Measure Results

The system compares the promoted group against a control group.

Campaign Result

Promoted Group Revenue Lift: +18.4%

Control Group Revenue Lift: +4.1%

Estimated True Incremental Lift: +14.3%

Repeat Purchase Lift: +9.6%

Gross Margin After Discount: 61%

---

9. How to Present This Without Showing Raw Data

The platform should show strategy, segments, and results without exposing customer records.

Show This

Recommended Segment

At-risk weekday customers

Segment Size

1,248 customers

Average Days Since Last Purchase

32 days

Recommended Low-Demand Window

Tuesday, 2 PM – 6 PM

Recommended Offer

$5 off $30+

Expected Incremental Revenue

$740

Expected Margin After Discount

58%

Confidence

86%

Avoid This

Customer Name: John Smith

Phone: 408-xxx-xxxx

Exact Purchase Record: Item X, Card Ending 1234

Presentation Principle

The UI should show:

- Segments, not raw customers
- Aggregates, not individual purchases
- Hashed or masked identifiers
- Confidence scores
- Margin impact
- Incremental lift
- Control group comparison
- Campaign recommendations
- Human approval checkpoints

---

10. Financial Factors to Account For

The model must protect the business from promotions that increase sales but reduce profitability.

Core Financial Factors

Factor	Why It Matters
Gross Sales	Measures total sales volume

Net Sales	Removes discounts, refunds, and taxes
Gross Margin	Shows profitability after product cost
Discount Cost	Measures the cost of the promotion
Redemption Rate	Shows how many customers used the offer
Incremental Revenue	Estimates revenue caused by the promotion
Cannibalization	Detects customers who would have bought anyway
Average Order Value	Measures transaction quality
Purchase Frequency	Measures retention behavior
Customer Lifetime Value	Captures long-term value
Cost of Goods Sold	Protects product-level margin
Payment Processing Fees	Captures transaction cost
Campaign Cost	Includes SMS/email/platform cost

Refund Rate	Detects quality or offer abuse issues
Inventory Impact	Prevents promoting unavailable or low-margin items

Most Important Formula

Promotion ROI =

(Incremental Gross Profit - Campaign Cost) / Campaign Cost

Incremental Profit Formula

Incremental Profit =

Incremental Revenue

- Discount Cost

- Cost of Goods Sold

- Payment Fees

- Campaign Cost

---

11. Flaws, Leaks, and Risks to Control

1. Over-Discounting Risk

The business may train customers to wait for discounts.

Controls:

- Limit promotion frequency per customer
- Use loyalty rewards instead of constant discounts
- Prioritize bundles over flat discounts
- Set minimum purchase thresholds

- Track discount dependency

---

2. Cannibalization Risk

Promotions may go to customers who would have purchased anyway.

Controls:

- Use control groups
- Use uplift modeling
- Compare promoted customers against similar non-promoted customers
- Track true incremental lift, not just redemption rate

---

3. Margin Leakage

Promotions may increase revenue while reducing profit.

Controls:

- Use product-level margin data
- Require minimum gross margin after discount
- Avoid discounts on already low-margin products
- Recommend bundles that protect margin

---

4. Customer Fatigue

Too many promotions may annoy customers.

Controls:

- Frequency caps
- Channel preference tracking
- Personalized offer timing
- Suppression rules for recently contacted customers
- Unsubscribe and preference management

---

5. Privacy and Data Leakage

Customer and POS data must remain protected.

Controls:

- Masked customer IDs
- No raw payment data storage
- Role-based access control
- Row-level security
- Export approval
- Audit logs
- Aggregated reporting by default

---

6. Bad Data and POS Sync Errors

Incorrect POS data can lead to incorrect campaigns.

Controls:

- Sync health monitoring
- Duplicate transaction detection
- Refund-to-sale matching
- Missing customer ID detection
- Schema drift alerts
- Daily reconciliation checks

---

7. Promotion Abuse

Customers or employees may exploit offers.

Controls:

- One-time-use promo codes
- Customer eligibility rules
- Employee override logs
- Refund-after-discount detection
- Duplicate redemption prevention
- Suspicious behavior alerts

---

12. Dashboard UI Structure

Main Navigation

1. Retention Dashboard
2. Low-Demand Calendar
3. Customer Segments
4. Promotion Recommendations
5. Campaign Builder
6. Margin Guardrails
7. A/B Test Results
8. Data Quality
9. Privacy & Access Control
10. Audit Trail

Key Dashboard Cards

Low-Demand Days Detected: 4

Recommended Campaigns: 7

Estimated Incremental Revenue: $3,420

Expected Gross Margin After Discount: 59%

At-Risk Customers: 1,248

Average Retention Lift: +9.6%

Cannibalization Risk: Medium

Data Quality Score: 94%

Recommended Visuals

Visual	Purpose
Low-Demand Calendar	Shows slow days and time windows
Customer Segment Cards	Shows who to target and why

Offer Recommendation Panel	Shows suggested campaign and expected ROI
Margin Guardrail Meter	Prevents unprofitable campaigns
Retention Cohort Chart	Tracks repeat behavior over time
A/B Test Result Panel	Shows true incremental lift
Campaign Audit Trail	Tracks who approved and launched campaigns

---

13. MVP Scope

MVP Features

1. POS data connection
2. Low-demand day detection
3. Customer segmentation
4. At-risk customer scoring
5. Promotion recommendation engine
6. Margin-safe offer rules
7. Campaign preview dashboard
8. Control group testing
9. Campaign performance report
10. Privacy-preserving aggregate views
11. Data-quality monitoring
12. Audit log

MVP Success Metrics

Metric	Target Direction

Repeat Purchase Rate	Increase
Low-Demand Day Revenue	Increase
Incremental Revenue	Increase
Gross Margin After Discount	Maintain or improve
Customer Reactivation Rate	Increase
Promotion Abuse	Decrease
Campaign ROI	Increase
Customer Fatigue	Decrease

---

14. Financial Modeling for the SMS-Only Retention Strategy

Business Concept

This strategy is a **low-demand day revenue and retention service** for small businesses that use a POS system. A local business, such as a pizza shop, can use the service to identify slow sales days, send targeted SMS promotions to opted-in customers, and increase repeat purchases without destroying margin.

The economic logic is:

> **The business sacrifices a controlled portion of short-term profit through a limited SMS offer in exchange for higher low-day traffic, more repeat visits, and stronger customer retention.**

This should be modeled as a **retention investment**, not just a discount campaign.

The Financial Model Should Answer

Will the profit from incremental orders and future repeat visits exceed the cost of discounts, SMS messages, payment fees, and the platform fee?

The model is successful only if the business owner can clearly see:

- How much extra revenue was created
- How much of that revenue was truly incremental
- How much profit was sacrificed through discounts
- Whether customers came back again after the promotion
- Whether the campaign improved low-demand days without hurting margins

---

Model Structure

POS System

↓

Historical Sales Baseline

↓

Low-Demand Day Detection

↓

SMS Opt-In Customer Segment

↓

Targeted Promotion

↓

Redemptions

↓

Incremental Orders

↓

Incremental Revenue

↓

Incremental Gross Profit

↓

Less: Discount Cost + SMS Cost + Payment Fees + Platform Fee

↓

Immediate Net Benefit

↓

Future Repeat Purchases

↓

90-Day Retention Value

POS Data Required

The POS system should provide the data needed to calculate baseline sales, customer behavior, order value, and campaign performance.

POS Data Field	Use in Financial Model
Transaction date and time	Identifies low-demand days and hours

Order total	Calculates average order value and campaign revenue
Discounts applied	Measures profit sacrificed
Refunds and voids	Adjusts true net sales
Product/category sold	Helps estimate gross margin and offer fit
Customer ID or phone hash	Tracks repeat behavior without exposing identity
Loyalty/SMS opt-in status	Determines who can legally receive campaigns
Store/location	Supports location-specific slow-day campaigns
Payment method and fees	Estimates transaction costs
Promo code/redemption tag	Attributes sales to the campaign

The model should work from **aggregated POS outputs**, not manually inspected raw customer records.

Core Financial Inputs

Input	Description

Baseline low-day revenue	Average revenue normally earned on the targeted slow day
Low-demand days per month	Number of campaign opportunities per month
SMS opt-in list size	Number of customers eligible to receive texts
Texts sent per campaign	Number of customers targeted per slow-day campaign
Average order value	Average POS ticket size
Gross margin	Revenue left after food/product cost
Net income margin	Final profit margin after operating expenses
Discount value	Dollar or percentage discount offered
Redemption rate	Percentage of recipients who use the offer
Incrementality rate	Percentage of redemptions that would not have happened without the SMS
Cannibalization rate	Percentage of redemptions from customers who would have bought anyway
SMS cost per message	Vendor/carrier cost per outbound SMS

Payment processing fee	Card/payment cost per transaction
Platform fee	What the service charges the small business
Retention conversion rate	Percentage of incremental customers who return again
Future repeat orders	Expected future orders from retained customers

---

Core Formulas

1. Monthly Texts Sent

Monthly Texts Sent = Low-Demand Days per Month × Texts Sent per Campaign

2. Redemptions

Redemptions = Monthly Texts Sent × Redemption Rate

3. Incremental Orders

Incremental Orders = Redemptions × Incrementality Rate

This is the key distinction. Redemptions are not automatically incremental. Some customers would have bought anyway.

4. Cannibalized Orders

Cannibalized Orders = Redemptions × Cannibalization Rate

5. Incremental Revenue

Incremental Revenue = Incremental Orders × Average Order Value

6. Incremental Gross Profit

Incremental Gross Profit = Incremental Revenue × Gross Margin

7. Discount Cost

Discount Cost = Redemptions × Average Discount Value

Discount cost applies to all redemptions, including customers who would have purchased anyway.

8. SMS Cost

SMS Cost = Monthly Texts Sent × SMS Cost per Message

9. Payment Processing Cost

Payment Processing Cost = Incremental Revenue × Payment Processing Rate

10. Immediate Campaign Profit

Immediate Campaign Profit =

Incremental Gross Profit

- Discount Cost

- SMS Cost

- Payment Processing Cost

- Platform Fee

11. Retained Customers Created

Retained Customers Created = Incremental Orders × Retention Conversion Rate

12. Future Retention Revenue

Future Retention Revenue =

Retained Customers Created × Future Repeat Orders × Future Average Order Value

13. Future Retention Gross Profit

Future Retention Gross Profit = Future Retention Revenue × Gross Margin

14. 90-Day Total Profit Impact

90-Day Total Profit Impact = Immediate Campaign Profit + Future Retention Gross Profit

15. Promotion ROI

Promotion ROI = 90-Day Total Profit Impact ÷ Promotion Investment

Where:

Promotion Investment = Discount Cost + SMS Cost + Platform Fee

---

Example Base Case: Local Pizza Shop

This example assumes a small pizza shop uses SMS promotions to lift sales on slower weekdays.

Metric	Example Value

Low-demand days per month	8
Texts sent per campaign	500
Monthly texts sent	4,000
SMS cost per text	$0.025
SMS cost	$100
Redemption rate	8%
Redemptions	320
Incrementality rate	60%
Incremental orders	192
Average order value	$28
Incremental revenue	$5,376
Gross margin	65%
Incremental gross profit	$3,494

Average discount value	$4
Discount cost	$1,280
Payment processing rate	2.9%
Payment processing cost	$156
Platform fee	$199
Immediate campaign profit	$1,859
Retention conversion rate	25%
Retained customers created	48
Future repeat orders per retained customer	1.5
Future retention revenue	$2,016
Future retention gross profit	$1,310
90-day total profit impact	$3,169

Base Case Interpretation

The campaign is financially attractive because the business gives up **$1,280 in discounts** plus campaign costs, but receives an estimated **$3,169 in 90-day profit impact**.

The important point is not that revenue increased. The important point is that **incremental gross profit and future retention value exceeded the cost of the promotion**.

Break-Even Analysis

The campaign breaks even when:

Incremental Gross Profit + Future Retention Gross Profit

≥ Discount Cost + SMS Cost + Payment Processing Cost + Platform Fee

The dashboard should always show a break-even redemption rate.

Break-Even Redemption Rate =

Fixed Campaign Costs ÷

[(Average Order Value × Gross Margin × Incrementality Rate) - Average Discount Value]

A campaign should not launch if the projected redemption rate is below the break-even redemption rate.

Sensitivity Cases

The model should show conservative, base, and aggressive outcomes so the business owner understands risk.

Case	Redemption Rate	Incrementality	90-Day Profit Impact	Interpretation
Conservative	4%	45%	Low / near break-even	Campaign may be too weak

Base	8%	60%	Positive	Campaign is financially attractive
Aggressive	12%	70%	Strong positive	Campaign should be repeated with guardrails

The two most important sensitivity drivers are:

1. **Incrementality** — did the SMS create new purchases?
2. **Gross margin after discount** — did the campaign remain profitable?

Client-Facing Dashboard Outputs

The product should avoid complex financial language and show the owner the simple business tradeoff.

Campaign: Tuesday Slow-Day Pizza Offer

Target Segment: At-risk weekday customers

Texts Sent: 4,000

Redemptions: 320

Estimated Incremental Orders: 192

Incremental Revenue: $5,376

Discount Cost: $1,280

SMS Cost: $100

Platform Fee: $199

Immediate Campaign Profit: $1,859

90-Day Retention Profit Impact: $3,169

Break-Even Redemption Rate: 4.3%

Client Net Benefit: Positive

Recommendation: Launch with control group

Business Model for the Service

The service itself can charge small businesses in three ways.

Pricing Model	Description	Pros	Cons
Flat Monthly Fee	Business pays one fixed subscription	Simple and predictable	Less upside for service provider
Performance Fee	Service takes a percentage of measured incremental profit	Strong alignment with client results	Requires trusted attribution
Hybrid Fee	Lower subscription plus performance fee	Best balance	Slightly more complex to explain

Recommended early pricing:

$199/month + 10% of measured incremental profit

Alternative simple MVP pricing:

$299/month including up to a set number of SMS campaigns

For small businesses, the pricing must feel obviously smaller than the expected profit lift.

Service Provider Unit Economics

The platform should also model its own profitability.

Service Provider Input	Example
Client monthly fee	$199
Performance fee	10% of incremental profit
Client incremental profit	$3,169
Performance revenue	$317
Total monthly revenue per client	$516
SMS cost covered by platform	$100
Gross profit before software/support costs	$416

This means the platform can be profitable if SMS costs, onboarding time, support, and POS integration costs stay controlled.

Margin Guardrails

A campaign should be blocked or flagged if any of these are true:

Expected gross margin after discount is too low.

Projected redemption rate is below break-even.

Incrementality estimate is weak.

The same customer segment has been promoted too often.

Discount cost exceeds expected incremental gross profit.

SMS unsubscribe risk is high.

The POS data quality score is too low.

Key Risks in the Financial Model

Risk	Why It Matters	Control
Cannibalization	Discounts go to customers who would have purchased anyway	Use holdout/control groups
Over-discounting	Customers learn to wait for offers	Use frequency caps and rotate offer types
Low-margin orders	Campaign increases sales but not profit	Use product/category margin rules
SMS fatigue	Customers unsubscribe or stop responding	Limit send frequency and personalize timing
Weak attribution	Business cannot prove the SMS worked	Use promo codes, POS tags, and holdout groups
Redemption abuse	Customers or employees exploit offers	One-time codes and redemption limits

Bad POS data	Incorrect data creates bad recommendations	Sync checks and reconciliation
Compliance risk	SMS requires proper opt-in and opt-out handling	Store consent records and automatic opt-out management

---

Best Financial Framing

This product should not be pitched as:

We help you send discounts.

It should be pitched as:

We help you turn slow days into repeat-customer revenue by using POS data to send margin-safe SMS offers to customers most likely to return.

The financial model must prove that the campaign creates **incremental profit**, not just temporary sales volume.

15. Final Product Philosophy

A weak promotion platform says:

Sales are slow. Send everyone a discount.

A strong retention platform says:

Demand is expected to be low Tuesday afternoon.

At-risk weekday customers are likely to respond.

A $5 off $30+ offer should improve retention while preserving margin.

Launch to a test group, hold back a control group, and measure true incremental lift.

The strongest product makes every recommendation explainable:

- Why this day?
- Why this customer segment?
- Why this offer?
- What is the expected revenue lift?
- What is the expected margin impact?
- What is the cannibalization risk?
- What proof do we have that the campaign worked?

That is how the product becomes more than a discount tool. It becomes a **retention optimization engine**.

Finance pt2

Pielot Pitch Script

SMS Retention Strategy for Pleasure Pizza Using POS Data

Opening

Pielot looks at your POS data, finds your slowest revenue windows, and automatically recommends SMS offers that bring the right customers back without destroying your margins.

For a restaurant like Pleasure Pizza, the goal is not to send more discounts. The goal is to make slow days more predictable by using customer behavior that already exists inside the POS system.

Pielot answers one simple question:

> Who should we text, when should we text them, what offer should we send, and will it actually make money?

Why Pleasure Pizza Is a Good Example

Pleasure Pizza is a strong example because it has the kind of local restaurant profile where repeat customers matter.

Publicly available information shows that Pleasure Pizza has a long-standing Santa Cruz presence, multiple customer-facing locations, dine-in, to-go, and delivery options, and online ordering connected to Toast for at least the Downtown location.

That matters because Pielot is designed for businesses that already have POS activity, repeat customers, and predictable demand swings.

Pleasure Pizza likely has natural revenue windows such as:

- Stronger weekend demand
- Beach and tourist traffic
- Lunch and dinner rushes
- Slower weekday afternoons

- Location-specific differences between Downtown, East Side, and Pleasure Point traffic

Pielot would use actual POS data to confirm these patterns instead of guessing.

The Problem

Restaurants do not struggle because every day is slow.

They struggle because demand is uneven.

Some windows are busy. Other windows are underused. Staff, kitchen capacity, and inventory are already available, but customer traffic is weaker than it could be.

Most restaurants respond with broad discounts.

That creates problems:

- Discounts may go to customers who would have ordered anyway
- Margins can shrink
- Customers may learn to wait for deals
- The owner may not know whether the campaign actually worked

Pielot solves this by making one clear recommendation at a time.

The Owner-Facing UI

The restaurant owner should not see a spreadsheet first.

The first screen should show five things:

1. Slow day detected
2. Recommended customer segment
3. Suggested offer
4. Expected profit
5. Launch / Edit / Reject

The finance model can live in the backend. The owner-facing experience should feel like an AI assistant making one clear recommendation.

Example Demo Screen

Tuesday Slow-Day Campaign

Slow window: Tuesday 2–6 PM
Target: At-risk weekday customers
Offer: $5 off $30+
Texts sent: 4,000
Expected redemptions: 320
Estimated new orders created: 192
Incremental revenue: $5,376
90-day profit impact: $3,169
Recommendation: Launch with test group

[Launch Campaign] [Edit Offer] [Reject]

This is the product.

Not a complicated dashboard.

A simple recommendation that the owner can understand in seconds.

Restaurant-Friendly Language

Pielot should avoid finance-heavy wording on the main screen.

Technical Term	Restaurant-Friendly Term
Incrementality rate	New orders created
Cannibalization risk	Would-have-ordered-anyway risk
Gross margin after discount	Profit after offer
Promotion ROI	Campaign profit return
Control group	Test comparison group

The detailed terms can still exist in the finance view, but the owner should see plain language.

How Pielot Works

Pielot connects to the POS system and looks for low-revenue patterns.

It uses data such as:

- Order date and time
- Ticket size
- Customer visit frequency
- Prior purchases
- Promo redemptions
- Discounts
- Refunds or voids
- Location
- SMS opt-in status

Then it recommends a campaign.

The workflow is:

POS Data
↓
Find slow revenue windows
↓
Pick the best customer segment
↓
Suggest a margin-safe SMS offer
↓
Estimate profit impact
↓
Owner approves, edits, or rejects
↓
Track actual results

What the Backend Calculates

The owner does not need to see every formula, but the backend needs to calculate them.

The most important financial logic is:

New Orders Created × Average Order Value = Incremental Revenue

Then:

Incremental Revenue
- Food/Product Cost
- Discount Cost
- SMS Cost
- Payment Fees
= Immediate Profit

Then Pielot adds future retention value:

Immediate Profit + Future Repeat Purchase Profit = 90-Day Profit Impact

The dashboard should show the result, not the entire spreadsheet.

Demo Walkthrough Script

Here is how I would explain the product in a presentation:

“Let’s say Pleasure Pizza usually has slower traffic on Tuesday afternoons from 2 to 6 PM. Pielot detects that slow window by looking at POS sales history.

Instead of telling the owner to discount everything, Pielot recommends a specific campaign.

For this example, the target is at-risk weekday customers — people who have ordered before during the week but have not come back recently.

The suggested offer is $5 off orders over $30.

Pielot estimates that if we send 4,000 texts, about 320 customers will redeem the offer. But the important number is not redemptions. The important number is new orders created.

Out of those redemptions, Pielot estimates 192 are new orders that likely would not have happened without the SMS.

That creates $5,376 in incremental revenue and an estimated 90-day profit impact of $3,169.

So the recommendation is simple: launch with a test comparison group.”

Why SMS Only

SMS keeps the product simple for local restaurants.

Restaurants do not need a complicated ad system. They need a fast way to reach customers when demand is weak.

SMS is useful because:

- It is immediate
- It is easy to understand
- It works for time-sensitive food offers
- It does not require customers to download an app
- It can connect directly to POS customer profiles and opt-in records

Pielot should feel like:

Slow day coming. Here is who to text. Here is the offer. Here is the expected profit. Launch?

Personalized and Group-Based Discounts

Pielot should not treat every customer the same.

Instead of sending one generic offer to the entire customer base, Pielot can tailor each discount based on the individual customer's POS behavior.

For example, if one customer usually visits on weekends and another customer usually visits on weekdays, they should not receive the same offer at the same time. Pielot should recommend the offer most likely to change each customer's behavior.

The model can look at:

- How much the customer usually spends
- Which products or categories they buy
- Which days and times they usually visit
- How often they return
- Whether they usually respond to discounts
- Whether they tend to order alone or with others
- Whether a larger offer could encourage a larger basket size

The goal is to incentivize additional spending, not just give away margin.

Individual Discount Example

Customer A
Usually visits: Friday evenings
Average spend: $18
Favorite item: Slices and drink
Suggested offer: $3 off $20+ on Tuesday afternoon
Goal: Move one visit into a slow window and increase basket size

Customer B
Usually visits: Sunday with family
Average spend: $42
Favorite item: Whole specialty pizza
Suggested offer: $5 off $35+ on Wednesday dinner
Goal: Bring back a higher-value customer during a weaker weekday

Friend and Group Discount Example

If the POS database frequently observes a few customers ordering together, Pielot can create group-based offers.

For example, if Sriram and Animesh often come in together or are part of the same ordering pattern, Pielot can recommend a group offer that encourages both people to redeem together.

Detected Pattern
Two customers often visit together on weekends.

Slow Window
Tuesday 2–6 PM

Suggested Group Offer
$8 off $45+ when both customers redeem together

Goal
Move a group visit into a slow window and increase total order value.

This makes the discount more strategic. The restaurant is not just rewarding one person. It is encouraging a larger group purchase during a low-demand window.

Why Group Discounts Matter

Group offers can increase:

- Total order size
- Visit frequency

- Customer retention
- Social purchasing behavior
- Low-demand day traffic

But Pielot should still protect margin. A group discount should only launch when the expected order size is high enough to justify the offer.

Why This Is Different

Most tools help restaurants send campaigns.

Pielot helps restaurants decide whether a campaign is worth sending and who should receive which offer.

A normal marketing tool says:

Send 10% off to your customer list.

Pielot says:

Tuesday 2–6 PM is projected to be slow.
At-risk weekday customers are the best target.
$5 off $30+ should create new orders while protecting profit.
Launch with a test comparison group.

That is the difference.

Pielot is not selling text messages.

Pielot is selling better slow-day revenue decisions.

What the Details View Shows

The full finance dashboard should be secondary.

The owner can click “View Details” to see:

- Texts sent
- Expected redemptions

- New orders created
- Would-have-ordered-anyway risk
- Incremental revenue
- Discount cost
- SMS cost
- Payment fees
- Profit after offer
- 90-day profit impact
- Campaign profit return
- Test comparison group results

This keeps the main UI simple while still giving financial transparency when needed.

Guardrails

Pielot should block or warn against bad campaigns.

A campaign should be flagged if:

- Profit after offer is too low
- Would-have-ordered-anyway risk is too high
- The customer segment has been texted too often
- The offer is too aggressive
- The SMS list has high unsubscribe risk
- The POS data is incomplete
- There is no test comparison group

This protects the restaurant from accidentally training customers to wait for discounts.

Closing

Pielot gives small restaurants a smarter way to use the data they already have.

For a business like Pleasure Pizza, the value is not just sending an SMS.

The value is knowing when demand is weak, which customers are worth contacting, what offer protects margin, and whether the campaign creates real profit.

The final pitch is:

> Pielot helps restaurants use POS data and SMS to turn slow days into profitable repeat-customer visits.

Not more random discounts.

Better timing. Better targeting. Better retention. Better slow-day revenue.

userflow

Here is the **simple UI-agnostic user flow** for **Pielot** — only screens, purpose, what each screen should include, and what it should generally feel like.

Pielot Simple User Flow

1. Landing / Welcome Screen

Purpose: Explain what Pielot does and get the restaurant owner started.

Should include:

- Product name: **Pielot**
- One-line explanation: “AI SMS revenue agent for local restaurants.”
- Clear value prop: “Find revenue opportunities and send personalized offers.”
- Main CTA: **Launch Revenue Agent**
- Secondary CTA: **Try Demo**
- 3 simple benefits:
    - Finds customers likely to order
    - Creates personalized SMS campaigns
    - Helps increase repeat revenue

Should feel like:
Simple, confident, business-focused. The user should instantly understand that Pielot helps restaurants make more money.

---

2. Restaurant Setup Screen

Purpose: Collect basic restaurant context so Pielot can personalize recommendations.

Should include:

- Restaurant name
- Cuisine type
- Location/time zone
- Average order value
- Popular items
- Slow days/hours
- Main goal:
    - Bring back old customers

    - Increase orders today
    - Promote a deal
    - Increase repeat customers
    - Fill slow hours
- Brand tone:
    - Friendly
    - Funny
    - Premium
    - Casual
    - Family-friendly

Primary CTA:
Continue

Should feel like:
A short setup form, not complicated onboarding.

---

3. Connect Data Screen

Purpose: Let the owner upload or connect customer/order data.

Should include:

- Upload CSV option
- Connect Google Sheet option
- Connect POS/CRM placeholder:
    - Toast
    - Square
    - Clover
    - DoorDash/Uber Eats exports
- Use demo data option
- Simple explanation of needed data:
    - Customer name
    - Phone number
    - Opt-in status
    - Last order date
    - Order count
    - Total spend
    - Favorite items

Primary CTA:
Upload Data or **Use Demo Data**

Should feel like:
Trustworthy and easy. The user should not feel like setup will take forever.

---

4. Data Validation Screen

Purpose: Confirm the data is usable and safe for SMS campaigns.

Should include:

- Total customers found
- Opted-in customers
- Excluded customers
- Invalid phone numbers
- Missing consent count
- Data quality score
- Preview table of customers
- Compliance status:
    - Opt-in required
    - Opt-outs excluded
    - Frequency caps enabled
    - Approval required before sending

Primary CTA:
Run Revenue Scan

Should feel like:
Reassuring. This screen should build trust and show that Pielot will not spam people.

---

5. AI Revenue Scan Screen

Purpose: Show Pielot analyzing the business and finding opportunities.

Should include:

- Loading/scanning state
- Agent activity feed:
    - Reading order history
    - Finding lapsed customers
    - Detecting high-value segments
    - Matching offers to behavior

    - Checking compliance
    - Ranking revenue opportunities
- Completion summary:
    - Customers analyzed
    - Opportunities found
    - Campaigns ready
    - Projected revenue

Primary CTA after scan:
View Opportunities

Should feel like:
Active and intelligent. This is where the user feels the agent is doing real work.

---

6. Main Dashboard Screen

Purpose: Show the owner the best revenue opportunities.

Should include:

- Top summary metrics:
    - Projected revenue
    - Customers analyzed
    - Opportunities found
    - Campaigns ready
- List of revenue opportunities:
    - Win back lapsed customers
    - Game-night bundle
    - Family Friday deal
    - Late-night student offer
    - Coupon-lover campaign
    - High-value customer upsell
- Each opportunity should show:
    - Audience size
    - Projected revenue
    - Priority
    - Recommended offer
    - Main action button

Primary actions:

- **Review Campaign**
- **Approve**

- **Edit**
- **Ignore**

Should feel like:
A command center. The owner should immediately see where money can be made.

---

7. Opportunity Detail Screen

Purpose: Explain one campaign recommendation in detail.

Should include:

- Opportunity title
  Example: **Win Back Lapsed Customers**
- Why Pielot found it
- Audience size
- Customer segment explanation
- Recommended offer
  Example: **Free garlic knots with any large pizza**
- Projected revenue
- Best send time
- Confidence score
- SMS preview
- Compliance checklist

Primary actions:

- **Approve Campaign**
- **Edit Message**
- **Change Offer**
- **Schedule Later**
- **Reject**

Should feel like:
Clear and persuasive. The owner should understand why this campaign is worth sending.

---

8. SMS Edit Screen

Purpose: Let the owner review or modify the generated message.

Should include:

- Generated SMS copy
- Character/SMS count
- Offer details
- Order link
- Opt-out line: “Reply STOP to opt out”
- Tone controls:
  - Make shorter
  - Make funnier
  - Make more direct
  - Make more premium
- Manual edit box
- Preview as customer would see it

Primary CTA:
Continue to Approval

Should feel like:
Simple and safe. The owner should feel in control.

---

9. Approval Screen

Purpose: Final review before scheduling or sending.

Should include:

- Campaign name
- Audience
- Recipient count
- SMS preview
- Offer
- Send time
- Projected revenue
- Compliance checklist:
  - Opt-in only
  - Opt-out included
  - Frequency cap passed
  - Quiet hours respected
  - Human approval required
- Confirmation checkboxes if needed

Primary actions:

- **Approve & Send**

- **Schedule Campaign**
- **Cancel**

Should feel like:
A final safety checkpoint. No accidental sending.

---

10. Campaign Scheduled / Sent Screen

Purpose: Confirm the campaign has been approved.

Should include:

- Success message
- Campaign status:
  - Scheduled
  - Sending
  - Sent
- Send time
- Recipient count
- Message preview
- Option to cancel if scheduled
- Link to performance page

Primary CTA:
View Performance

Should feel like:
Clean confirmation. The user should know exactly what happened.

---

11. Performance Screen

Purpose: Show how the campaign performed.

Should include:

- Sent count
- Delivered count
- Clicks
- Replies
- Orders
- Revenue generated

- Opt-outs
- Average order value
- ROI
- Best-performing segment
- Pielot learning insights

Example insights:

- “Garlic knots offers performed best.”
- “Best response time was 4–6 PM.”
- “Lapsed customers responded better after 30 days than 60 days.”

Primary actions:

- **Create Similar Campaign**
- **Run New Scan**
- **View Full Report**

Should feel like:
Proof of value. The owner should clearly see whether Pielot made them money.

Secondary Screens

12. Campaigns Screen

Purpose: Manage all campaigns.

Should include:

- Draft campaigns
- Scheduled campaigns
- Active campaigns
- Completed campaigns
- Rejected campaigns

Each campaign should show:

- Name
- Audience
- Status
- Send time
- Projected revenue

- Actual revenue
- Actions

---

13. Customers Screen

Purpose: Browse customer data and segments.

Should include:

- Customer list
- Search/filter
- Opt-in status
- Last order date
- Total spend
- Favorite item
- Segment
- Last campaign received

---

14. Customer Profile Screen

Purpose: Show individual customer behavior.

Should include:

- Name
- Phone
- Opt-in status
- Last order
- Order count
- Total spend
- Favorite items
- Preferred order time
- Campaign history
- Recommended next offer

---

15. Segments Screen

Purpose: Show AI-created customer groups.

Should include segments like:

- Lapsed customers
- Friday regulars
- Family buyers
- Coupon lovers
- Game-night buyers
- Late-night customers
- High-value customers
- One-time customers

Each segment should show:

- Customer count
- Average spend
- Revenue potential
- Suggested campaign

---

16. Agent Chat Screen

Purpose: Let the owner ask Pielot what to do.

Should include:

- Chat interface
- Suggested prompts:
  - “What campaign should I send today?”
  - “How do I increase orders tonight?”
  - “Find customers likely to reorder.”
  - “Make this offer less discount-heavy.”
- Campaign preview beside the chat
- Approval required before sending

Should feel like:
A business assistant, not a generic chatbot.

---

17. Compliance Screen

Purpose: Manage SMS safety and legal guardrails.

Should include:

- Opted-in customer count
- Opted-out customer list
- Missing consent list
- Frequency cap settings
- Quiet hours
- Approval rules
- Campaign audit log
- STOP handling status

Should feel like:
Transparent and trustworthy.

---

18. Settings Screen

Purpose: Manage restaurant and account settings.

Should include:

- Restaurant profile
- Brand tone
- Popular items
- Offers allowed
- Max discount
- SMS send rules
- Team members
- Integrations
- Billing

Main Happy Path

This is the simplest flow to build first:

Landing
→ Restaurant Setup
→ Connect Data
→ Data Validation
→ AI Revenue Scan
→ Dashboard
→ Opportunity Detail

→ SMS Edit
→ Approval
→ Campaign Sent/Scheduled
→ Performance

Hackathon MVP Flow

For the hackathon, build only this:

Landing
→ Use Demo Data
→ AI Revenue Scan
→ Dashboard
→ Opportunity Detail
→ Approval
→ Results

That is enough to make Pielot feel complete.

Minimum Screens to Build First

1. Landing
2. Demo Data / Connect Data
3. AI Revenue Scan
4. Dashboard
5. Opportunity Detail
6. Approval
7. Results

These 7 screens are enough for a strong demo.

detiaeld prd

Pielot User Flow + Screens

Product

Pielot
AI SMS revenue agent for local restaurants.

Core User Flow Summary

Pielot's main flow is:

1. Restaurant owner lands on Pielot.
2. Owner creates restaurant profile.
3. Owner uploads customer/order data or connects a source.
4. Pielot validates opt-in and customer data.
5. Pielot runs an AI revenue scan.
6. Pielot finds revenue opportunities.
7. Owner reviews recommended SMS campaigns.
8. Owner approves, edits, schedules, or rejects campaigns.
9. Pielot sends or simulates the campaign.
10. Owner tracks campaign performance.
11. Pielot learns and recommends the next campaign.

---

1. Public Landing Page

Goal

Explain what Pielot does in 5 seconds and push the user to launch the revenue agent.

URL

`/`

Hero Section

Main headline

AI SMS revenue agent for local restaurants

Subheadline

Pielot analyzes customer behavior, finds hidden revenue opportunities, and creates personalized SMS campaigns that bring customers back.

Primary CTA

Launch Revenue Agent

Secondary CTA

View Demo

Hero Visual

A mock dashboard showing:

- 412 customers analyzed
- 6 revenue opportunities found
- $3,240 projected revenue
- 3 campaigns ready for approval

Sections

Section 1: How It Works

Three steps:

1. **Connect your customer data**
   Upload a CSV or connect your POS/customer list.
2. **Pielot finds revenue opportunities**
   It detects lapsed customers, upsell moments, slow-hour gaps, and repeat-order patterns.
3. **Approve campaigns**
   Pielot writes personalized SMS offers and waits for your approval before sending.

Section 2: Example Campaigns

Cards:

- Win back lapsed customers
- Fill slow lunch hours
- Launch game-night bundles
- Upsell regulars
- Bring back abandoned orders
- Ask loyal customers for referrals

Section 3: Compliance-Safe Messaging

Show trust badges:

- Opt-in only
- STOP opt-out handling
- Frequency caps
- Human approval before send
- Campaign audit logs

Section 4: Restaurant Use Cases

Use-case cards:

- Pizza shops
- Cafes
- Food trucks
- Burger shops
- Indian restaurants
- Small chains

Final CTA

Find Revenue Opportunities

---

2. Sign Up / Login Screen

URL

`/auth`

Goal

Let restaurant owners create an account quickly.

Components

Left Side

Brand panel:

Pielot
Your AI pilot for restaurant revenue.

Small animation:

- Customer data goes in
- Revenue campaigns come out

Right Side

Auth card:

Fields:

- Email
- Password

Buttons:

- Continue with Email
- Continue with Google

Links:

- Already have an account? Log in
- New here? Create account

Post-login redirect

New users go to:

`/onboarding/restaurant`

Returning users go to:

`/dashboard`

---

3. Restaurant Profile Setup

URL

`/onboarding/restaurant`

Goal

Collect restaurant context so the AI can create relevant campaigns.

Screen Title

Tell Pielot about your restaurant

Fields

Basic Info

- Restaurant name
- Cuisine type
- City
- Time zone
- Website/order link
- Phone number

Restaurant Type

Dropdown:

- Pizza
- Cafe
- Burger
- Indian
- Mexican
- Food truck
- Bakery
- Other

Business Goals

Checkboxes:

- Bring back old customers
- Increase weekday orders
- Increase lunch orders
- Increase dinner orders
- Promote new menu items
- Increase average order value
- Get more repeat customers
- Get more reviews/referrals

Popular Items

Multi-select or free text:

- Pepperoni pizza
- Wings
- Garlic knots
- Family combo
- Veggie pizza
- Lunch slice deal
- Soda

Brand Tone

Options:

- Friendly
- Funny
- Premium
- Casual
- Gen Z
- Family-friendly
- Short and direct

Average Order Value

Input:

- $24

Slow Days

Checkboxes:

- Monday
- Tuesday
- Wednesday
- Thursday
- Friday
- Saturday
- Sunday

Slow Hours

Time range input:

- 2 PM–5 PM
- 9 PM–11 PM
- 11 AM–1 PM

Buttons

- Back
- Save & Continue

Next screen

`/onboarding/data-source`

---

4. Data Source Selection Screen

URL

`/onboarding/data-source`

Goal

Let the owner upload or connect customer/order data.

Screen Title

Connect your customer data

Subtitle

Pielot uses your customer and order history to find revenue opportunities.

Source Cards

Card 1: Upload CSV

Best for hackathon MVP.

Button:

Upload CSV

Card 2: Google Sheets

Button:

Connect Sheet

Card 3: POS Export

Display as coming soon or mock integrations:

- Toast
- Square
- Clover
- DoorDash
- Uber Eats

Button:

Coming Soon

Card 4: Use Demo Data

For hackathon demo.

Button:

Use Pleasure Pizza Demo Data

Required Data Explanation

Show small checklist:

- Customer name
- Phone number
- SMS opt-in status
- Last order date
- Order count
- Total spend
- Favorite items
- Coupon usage
- Preferred order time

Buttons

- Back
- Continue

Next screen

If CSV selected:

`/onboarding/upload`

If demo data selected:

`/agent/scan`

---

5. CSV Upload Screen

URL

`/onboarding/upload`

Goal

Upload customer/order data and validate file quality.

Screen Title

Upload your customer list

Components

Upload Box

Drag-and-drop CSV upload.

Accepted formats:

- `.csv`
- `.xlsx`

CSV Template Button

Download sample template

Required Columns Checklist

- customer_id
- first_name
- phone_number
- sms_opt_in_status
- last_order_date
- order_count
- total_spend
- average_order_value
- favorite_items

Optional Columns

- preferred_order_day
- preferred_order_time
- coupon_usage_score
- delivery_or_pickup
- last_campaign_received
- loyalty_points

Upload States

Empty State

“Drop your customer CSV here.”

Uploading State

Progress bar.

Success State

“412 customers found.”

Error State

“Missing required column: sms_opt_in_status.”

Buttons

- Back
- Validate Data

Next screen

`/onboarding/data-validation`

---

6. Data Validation Screen

URL

`/onboarding/data-validation`

Goal

Show whether the data is usable and safe for SMS campaigns.

Screen Title

Data check complete

Top Summary Cards

Card 1

412 customers found

Card 2

387 opted-in customers

Card 3

25 excluded

Reason: missing consent or opted out

Card 4

92% data quality score

Validation Sections

Section 1: Consent Check

Rows:

- Opted-in customers: 387
- Opted-out customers: 14
- Missing consent status: 11
- Invalid phone numbers: 6

Status:

Ready with exclusions

Section 2: Order History Check

Rows:

- Customers with order history: 399
- Customers missing order history: 13
- Average order value detected: $24.30

- Most popular item: Pepperoni pizza

Section 3: Campaign Safety

Rows:

- Opt-out handling: enabled
- Frequency caps: enabled
- Human approval: required
- Sensitive targeting: disabled

Table Preview

Show first 10 customer rows:

Name	Phone	Opt-in	Last Order	Orders	Spend	Favorite
Ravi	***-4321	Yes	4 days ago	12	$284	Pepperoni
Maya	***-8821	Yes	63 days ago	5	$110	Veggie
Alex	***-1901	Yes	9 days ago	8	$176	Wings

Buttons

- Back
- Fix Data
- Run Revenue Scan

Next screen

`/agent/scan`

---

7. AI Revenue Scan Screen

URL

`/agent/scan`

Goal

Make the AI agent feel alive and show that Pielot is doing actual analysis.

Screen Title

Pielot is scanning for revenue opportunities

Main Visual

Animated agent workflow.

Example steps:

1. Reading customer order history
2. Finding repeat purchase patterns
3. Detecting lapsed customers
4. Finding slow-hour opportunities
5. Matching offers to customer behavior
6. Writing SMS campaign options
7. Checking opt-in and frequency caps
8. Ranking campaigns by projected revenue

Progress UI

Progress bar from 0–100%.

Agent Log

Live feed:

- Data Ingestion Agent cleaned 412 customer records.
- Segmentation Agent found 8 customer groups.
- Opportunity Agent detected 6 revenue opportunities.
- Offer Agent generated 12 offer ideas.
- SMS Agent wrote 9 campaign drafts.
- Compliance Agent excluded 25 unsafe recipients.

- Revenue Agent ranked 4 campaigns as high priority.

Completion State

When scan completes:

6 revenue opportunities found
$3,240 projected revenue
3 campaigns ready for approval

Button

View Opportunities

Next screen

`/dashboard`

---

8. Main Dashboard

URL

`/dashboard`

Goal

Show the owner exactly where money can be made.

Header

Left:

Pielot Dashboard

Right:

- Restaurant selector

- Date range
- Run New Scan
- Settings

Top KPI Cards

Card 1

$3,240
Projected revenue opportunities

Card 2

412
Customers analyzed

Card 3

6
Opportunities found

Card 4

3
Campaigns ready for approval

Card 5

387
Opted-in customers

Main Layout

Left Column: Revenue Opportunities

Cards ranked by priority.

Each card shows:

- Campaign name
- Audience
- Projected revenue
- Confidence score

- Risk level
- Recommended action

Example cards:

Win Back Lapsed Customers

143 customers
Projected revenue: $860
Action: Free garlic knots offer
Priority: High

Game Night Bundle

87 customers
Projected revenue: $1,120
Action: Pizza + wings bundle
Priority: High

Family Friday Deal

64 customers
Projected revenue: $740
Action: 2 pizzas + soda
Priority: Medium

Late Night Student Deal

118 customers
Projected revenue: $520
Action: Slice + drink deal
Priority: Medium

Center Column: Campaign Preview

When a campaign card is selected, show:

- Segment explanation
- Offer recommendation
- SMS draft
- Send time
- Projected results
- Compliance status

Right Column: Agent Activity

Show latest logs:

- “Found 143 lapsed customers”
- “Selected free garlic knots instead of 20% off to protect margin”
- “Excluded 14 opted-out contacts”
- “Recommended send time: 4:30 PM”

Buttons

- Review Campaign
- Approve
- Edit
- Schedule
- Ignore

Navigation Sidebar

Items:

- Dashboard
- Revenue Scan
- Campaigns
- Customers
- Segments
- Performance
- Compliance
- Settings

---

9. Opportunity Detail Screen

URL

`/opportunities/:id`

Goal

Explain why an opportunity exists and what Pielot recommends doing.

Screen Title

Example:

Win Back Lapsed Customers

Top Summary

- Audience: 143 customers
- Projected revenue: $860
- Recommended offer: Free garlic knots
- Best send time: Today at 4:30 PM
- Confidence: 82%
- Status: Ready for approval

Section 1: Why Pielot Found This

Plain-English explanation:

“143 customers have ordered from Pleasure Pizza before but have not ordered in 45+ days. Many of them previously responded to side-item offers. A free garlic knots offer is likely to bring them back without heavily discounting pizza.”

Section 2: Customer Segment

Table:

Segment	Count	Avg Spend	Last Order	Favorite Items
Lapsed customers	143	$22	45+ days ago	Pepperoni, Garlic Knots

Section 3: Offer Strategy

Show:

- Chosen offer: Free garlic knots with large pizza
- Reason: Low food cost, high perceived value
- Alternative offers:
    - 10% off
    - Free soda

  - Free delivery
- Why alternatives were not selected:
  - 10% off hurts margin more
  - Free soda has lower excitement
  - Free delivery only works for delivery customers

Section 4: Generated SMS

Message box:

“Pleasure Pizza: We miss you. Get free garlic knots with any large pizza tonight. Order here: [link]. Reply STOP to opt out.”

Actions:

- Regenerate
- Make funnier
- Make shorter
- Make more premium
- Edit manually

Section 5: Revenue Projection

Metrics:

- Recipients: 143
- Expected conversion: 12%
- Expected orders: 17
- Average order value: $24
- Projected revenue: $408
- Upside estimate: $860
- Estimated gross profit: $290

Section 6: Compliance Check

Checklist:

- Opt-in customers only
- Opted-out customers excluded
- STOP language included
- Frequency cap passed
- Human approval required
- No sensitive targeting

- Send time allowed

Buttons

- Approve Campaign
- Schedule Campaign
- Edit Campaign
- Reject Opportunity
- Back to Dashboard

---

10. Campaign Approval Modal

Trigger

User clicks **Approve Campaign**.

Goal

Final safety and confirmation step before sending.

Modal Title

Approve SMS campaign?

Campaign Summary

- Campaign: Win Back Lapsed Customers
- Audience: 143 customers
- Send time: Today, 4:30 PM
- Offer: Free garlic knots with large pizza
- Estimated revenue: $860

SMS Preview

"Pleasure Pizza: We miss you. Get free garlic knots with any large pizza tonight. Order here: [link]. Reply STOP to opt out."

Required Confirmation

Checkboxes:

- I confirm these customers opted in to SMS marketing.
- I confirm this offer is accurate.
- I understand customers can opt out by replying STOP.

Buttons

- Cancel
- Schedule Instead
- Approve & Send

Post-approval state

Show success toast:

Campaign approved. Pielot scheduled it for 4:30 PM.

Next screen

`/campaigns/:id`

---

11. Campaigns List Screen

URL

`/campaigns`

Goal

Show all draft, scheduled, active, and completed campaigns.

Screen Title

Campaigns

Tabs

- Drafts
- Scheduled
- Active
- Completed
- Rejected

Campaign Table

Columns:

- Campaign
- Segment
- Recipients
- Status
- Send time
- Projected revenue
- Actual revenue
- Opt-outs
- Actions

Example rows:

Campaign	Segment	Recipients	Status	Projected	Actual
Win Back Lapsed Customers	Lapsed	143	Scheduled	$860	—
Game Night Bundle	Sports buyers	87	Draft	$1,120	—
Family Friday	Families	64	Completed	$740	$812

Actions

- View
- Edit
- Duplicate
- Cancel
- View Results

---

12. Campaign Detail / Sending Screen

URL

`/campaigns/:id`

Goal

Show live campaign status and performance.

States

Draft State

Show:

- Campaign details
- SMS copy
- Audience
- Approval checklist
- Approve button

Scheduled State

Show:

- Campaign scheduled time
- Countdown
- Audience locked
- Cancel button
- Edit before send button

Sending State

Show:

- Sending progress
- Messages sent
- Delivered

- Failed
- Opt-outs

Completed State

Show:

- Campaign results
- Revenue attributed
- Orders generated
- Lessons learned

Metrics

- Sent
- Delivered
- Clicked
- Replied
- Ordered
- Revenue
- Opt-outs
- ROI

Agent Insights

Example:

“Customers who previously ordered garlic knots converted 1.8x better than customers who had only ordered pizza. Pielot recommends using side-item offers again for this segment.”

---

13. Campaign Performance Screen

URL

`/performance`

Goal

Show how much revenue Pielot has generated.

Top KPI Cards

- Total revenue generated
- Orders attributed
- Revenue per SMS
- Repeat customers recovered
- Best campaign
- Opt-out rate

Charts

Chart 1: Revenue by Campaign

Bar chart.

Chart 2: Orders Over Time

Line chart.

Chart 3: Segment Performance

Table or chart:

- Lapsed customers
- Friday regulars
- Game-night buyers
- Family buyers
- Late-night customers

Chart 4: Offer Performance

Compare:

- Free garlic knots
- 10% off
- Free drink
- Family combo
- Game-night bundle

AI Summary

Pielot writes:

“Your best-performing campaign this month was Game Night Bundle, generating $1,120 from 87 recipients. Customers who order after 6 PM respond best to bundled offers, while lapsed customers respond better to free side items than percentage discounts.”

Buttons

- Run New Scan
- Create Similar Campaign
- Export Report

---

14. Customers Screen

URL

`/customers`

Goal

Let restaurant owners see customer segments and individual profiles.

Screen Title

Customers

Top Filters

- Search by name/phone
- Segment
- Opt-in status
- Last order date
- Favorite item
- Total spend
- Campaign received
- Ordered after campaign

Customer Table

Columns:

- Name
- Phone
- Opt-in
- Last order
- Orders
- Total spend
- Favorite item
- Segment
- Last campaign

Example:

Name	Opt-in	Last Order	Orders	Spend	Segment
Ravi	Yes	4 days ago	12	$284	Friday Regular
Maya	Yes	63 days ago	5	$110	Lapsed
Alex	Yes	9 days ago	8	$176	Coupon Lover

Actions

- View profile
- Add to segment
- Exclude from campaigns
- Mark opted out
- Export

---

15. Customer Profile Screen

URL

`/customers/:id`

Goal

Show customer-level intelligence.

Sections

Customer Summary

- Name
- Phone
- Opt-in status
- Total spend
- Order count
- Average order value
- Last order date
- Favorite item
- Preferred order day/time

Buying Pattern

Example:

“Ravi usually orders pepperoni pizza on Friday evenings and has responded to bundle offers twice.”

Campaign History

Table:

- Campaign
- Date
- Message
- Clicked
- Ordered
- Revenue

Recommended Next Action

Example:

“Send Ravi a Friday pepperoni bundle offer at 4:30 PM.”

Buttons:

- Generate SMS
- Add to Campaign
- Exclude
- Mark Opted Out

---

16. Segments Screen

URL

`/segments`

Goal

Show all AI-generated customer groups.

Segment Cards

Each card includes:

- Segment name
- Customer count
- Average order value
- Revenue potential
- Recommended campaign
- Last contacted

Segments:

1. Lapsed Customers
2. Friday Regulars
3. Game-Night Buyers
4. Family Buyers
5. Coupon Lovers
6. High-Value Regulars
7. Lunch Crowd
8. Late-Night Customers
9. New One-Time Customers
10. Abandoned Checkout Customers

Segment Detail

Clicking a segment shows:

- Segment rules
- Customer list
- Suggested offers
- Past performance
- Recommended send time
- Create campaign button

---

17. Create Campaign Screen

URL

`/campaigns/new`

Goal

Allow manual campaign creation with AI assistance.

Step 1: Choose Goal

Options:

- Bring back lapsed customers
- Increase orders today
- Promote a menu item
- Fill slow hours
- Increase average order value
- Get reviews/referrals
- Custom goal

Step 2: Choose Audience

Options:

- AI recommended segment

- Manual segment
- Upload list
- All opted-in customers

Step 3: Choose Offer

Options:

- AI recommended offer
- Free item
- Percentage discount
- Bundle
- Limited-time deal
- Custom offer

Step 4: Generate SMS

Pielot generates 3 variants:

- Friendly
- Urgent
- Funny

Owner selects or edits.

Step 5: Review & Approve

Show:

- Audience
- Message
- Offer
- Send time
- Compliance checklist

Buttons

- Save Draft
- Schedule
- Approve & Send

---

18. Agent Chat Screen

URL

`/agent`

Goal

Let the owner talk to Pielot naturally.

Chat Examples

Owner asks:

“What should I send today to make more money?”

Pielot replies:

“I found 3 strong campaigns for today: a win-back offer for 143 lapsed customers, a family dinner deal for 64 high-AOV customers, and a late-night slice deal for 118 customers. The win-back offer has the highest projected ROI.”

Owner asks:

“Make it less discount-heavy.”

Pielot replies:

“Instead of 20% off, I recommend free garlic knots with a large pizza. It protects margin while still creating perceived value.”

Owner asks:

“Schedule the best one for 4 PM.”

Pielot replies:

“I can schedule the Win Back Lapsed Customers campaign for 4 PM. Please approve the final message before sending.”

Chat UI

Left:

- Chat messages

Right:

- Campaign preview card
- Segment preview
- Approval button

Guardrail

Pielot never sends directly from chat without explicit campaign approval.

---

19. Compliance Center Screen

URL

`/compliance`

Goal

Make the owner trust Pielot and prevent unsafe campaigns.

Sections

Consent Overview

- Total customers
- Opted-in
- Opted-out
- Missing consent
- Invalid phone numbers

Opt-Out List

Table:

- Phone number

- Keyword
- Date
- Campaign source

Frequency Caps

Settings:

- Max texts per week
- Max texts per month
- Quiet hours
- Minimum days between campaigns

Message Rules

Checklist:

- Include restaurant name
- Include opt-out language
- No misleading urgency
- No sensitive targeting
- No unapproved sends

Campaign Audit Log

Rows:

- Campaign
- Approved by
- Approval time
- Recipient count
- Compliance status

---

20. Settings Screen

URL

`/settings`

Goal

Manage restaurant, brand, integrations, and campaign rules.

Tabs

Restaurant Profile

- Name
- Cuisine
- Address
- Time zone
- Order link
- Average order value
- Popular items

Brand Voice

- Friendly
- Funny
- Premium
- Casual
- Family-friendly
- Custom instructions

Example custom instruction:

"Keep messages short, playful, and never too pushy."

Campaign Rules

- Max discount
- Minimum margin
- Allowed offers
- Approval required
- Send time window
- Frequency caps

Integrations

- CSV
- Google Sheets
- Twilio
- Toast
- Square
- Clover

Team

- Owner
- Manager
- Marketing user
- Roles and permissions

Billing

- Plan
- Usage
- SMS credits
- Payment method

---

21. Empty States

Empty Dashboard

If no data uploaded:

No customer data yet

Subtext:

Upload your customer list so Pielot can find revenue opportunities.

Button:

Upload Data

No Opportunities Found

No strong campaigns found today

Subtext:

Pielot did not find a campaign that meets your revenue and compliance thresholds. Try lowering the minimum audience size or uploading more order history.

Button:

Adjust Settings

No Campaigns Yet

No campaigns created

Subtext:

Run your first revenue scan to generate AI campaign ideas.

Button:

Run Revenue Scan

---

22. Error States

Missing Consent Data

Title:

Consent status missing

Message:

Pielot cannot send SMS campaigns unless customer opt-in status is included.

Actions:

- Upload corrected file
- Download CSV template
- Continue with demo data

Too Many Opted-Out Customers

Title:

Limited reachable audience

Message:

Only 42 customers are currently eligible for SMS campaigns.

Actions:

- View eligible customers
- Import more opted-in customers

SMS Copy Failed Compliance

Title:

Message needs review

Message:

This SMS is missing opt-out language or contains unsupported urgency.

Actions:

- Auto-fix message
- Edit manually

---

23. Main Happy Path

This is the ideal demo flow.

1. User lands on homepage.
2. Clicks **Launch Revenue Agent**.
3. Creates account or enters demo.
4. Creates restaurant profile: Pleasure Pizza.
5. Selects **Use Demo Data**.
6. Pielot scans 412 customers.
7. Dashboard shows 6 revenue opportunities.
8. User clicks **Win Back Lapsed Customers**.
9. Pielot explains the segment and offer.
10. User reviews SMS.
11. Compliance checklist passes.
12. User clicks **Approve Campaign**.
13. Campaign is scheduled.
14. Performance screen shows simulated results.

---

24. Hackathon Demo Flow

For hackathon, reduce the flow to 5 polished screens:

Screen 1: Landing

Show Pielot value proposition.

CTA:

Try with Pleasure Pizza

Screen 2: Agent Scan

Show Pielot analyzing 412 customers.

Screen 3: Dashboard

Show:

- $3,240 projected revenue
- 6 opportunities found
- 3 campaigns ready

Screen 4: Campaign Detail

Show:

- Win-back campaign
- SMS copy
- projected revenue
- compliance checklist

Screen 5: Results

Show:

- campaign sent/scheduled
- projected orders
- simulated performance
- agent learning

---

25. Mobile Flow

Since restaurant owners may use phones, Pielot should feel mobile-friendly.

Mobile Navigation

Bottom nav:

- Home
- Campaigns
- Agent
- Customers
- Settings

Mobile Dashboard Cards

Stacked cards:

1. Projected revenue
2. Campaigns ready
3. Top opportunity
4. Approve campaign button

Mobile Approval Flow

Owner can approve campaign in 3 taps:

1. Open campaign
2. Review SMS
3. Tap Approve

---

26. Permissions

Owner

Can:

- Upload data
- Approve campaigns
- Send campaigns
- Change settings
- Manage billing

Manager

Can:

- View dashboard
- Create campaigns
- Edit SMS
- Schedule drafts

Cannot:

- Send without owner approval unless allowed

Viewer

Can:

- View dashboard
- View reports

Cannot:

- Edit or approve campaigns

---

27. Notification Flows

Daily Opportunity Notification

Message:

“Pielot found 3 campaigns that could generate $1,420 today. Review now.”

Campaign Approval Reminder

Message:

“Your Family Friday campaign is ready for approval before 4 PM.”

Performance Update

Message:

“Game Night Bundle generated 28 orders and $712 in tracked revenue.”

Compliance Alert

Message:

“Opt-out rate was higher than usual. Pielot recommends lowering frequency this week.”

---

28. Final Screen List

Public

1. Landing Page
2. Pricing Page
3. Demo Page
4. Login / Signup

Onboarding

5. Restaurant Profile Setup
6. Data Source Selection
7. CSV Upload
8. Data Validation
9. Consent Confirmation
10. First Revenue Scan

Core App

11. Dashboard
12. Revenue Scan
13. Opportunity Detail
14. Campaign Approval Modal
15. Campaigns List
16. Campaign Detail
17. Campaign Performance
18. Customers
19. Customer Profile
20. Segments
21. Segment Detail
22. Create Campaign
23. Agent Chat
24. Compliance Center
25. Settings

Admin / Future

26. Team Management
27. Billing
28. Integrations
29. Reports
30. Multi-Location Dashboard

---

29. Recommended Hackathon Scope

Build these screens first:

1. Landing page
2. Demo data selection
3. Agent scan animation
4. Main dashboard
5. Campaign detail page
6. Approval modal
7. Campaign performance page

Do not spend time on:

- Real auth
- Real billing
- Full POS integrations

- Full customer table
- Multi-location support
- Deep analytics

---

30. Core UX Principle

Every screen should answer one of these questions:

1. **Where is the money?**
2. **Who should we message?**
3. **What should we offer?**
4. **Why will this work?**
5. **Is it safe to send?**
6. **How much did it make?**

If a screen does not answer one of those, it should not be in the MVP.

---

31. Final Product Flow in One Sentence

A restaurant owner opens Pielot, connects customer data, lets the AI scan for revenue opportunities, reviews personalized SMS campaigns, approves the best one, and tracks how many orders it generated.

Tab 6

Pielot Pitch Script

SMS Retention Strategy for Pleasure Pizza Using POS Data

Opening

Pielot looks at your POS data, finds your slowest revenue windows, and automatically recommends SMS offers that bring the right customers back without destroying your margins.

For a restaurant like Pleasure Pizza, the goal is not to send more discounts. The goal is to make slow days more predictable by using customer behavior that already exists inside the POS system.

Pielot answers one simple question:

> Who should we text, when should we text them, what offer should we send, and will it actually make money?

Website-Ready Positioning

One-Line Pitch

Pielot helps restaurants turn slow days into profitable repeat visits by using POS data to recommend personalized SMS offers that protect margins.

Simple Hero Message

Your POS already knows when revenue slows down.
Pielot tells you who to text, what offer to send, and whether the campaign should make money.

What the Website Should Communicate Immediately

The website should make the product feel simple, not technical.

A restaurant owner should understand Pielot in under ten seconds:

1. Pielot connects to your POS.
2. It finds your slowest revenue windows.
3. It recommends personalized SMS offers.
4. It estimates profit before you launch.
5. You approve, edit, or reject the campaign.

The main screen should not look like a spreadsheet. It should look like one clear AI recommendation.

Website Financial Model Experience

The website should not present the financial model like a spreadsheet.

The financial model should work quietly in the background and appear to the restaurant owner as a simple recommendation.

The owner should feel like Pielot is saying:

We found a slow revenue window.
We found the customer group most likely to come back.
We created an offer that should protect profit.
Here is the expected result.
Do you want to launch it?

Primary Website View

The first screen should only show five things:

1. Slow day detected
2. Recommended customer segment
3. Suggested offer
4. Expected profit
5. Launch / Edit / Reject

Everything else should live behind a “View Details” button.

The goal is to make Pielot feel like an AI assistant for restaurant revenue decisions, not a finance dashboard.

Owner-Facing Recommendation Card

This should be the center of the website demo.

Tuesday Slow-Day Campaign

Slow window: Tuesday 2–6 PM
Target: At-risk weekday customers
Offer: $5 off $30+
Texts sent: 4,000
Expected redemptions: 320
Estimated new orders created: 192
Incremental revenue: $5,376
90-day profit impact: $3,169
Recommendation: Launch with test group

[Launch Campaign] [Edit Offer] [Reject]

Why This Works

This card gives the owner the full business decision without forcing them to understand the full model.

The card answers:

- When is revenue slow?
- Who should we target?
- What offer should we send?
- How much revenue could it create?
- How much profit could it create over time?
- Should we launch, edit, or reject?

That is the correct website experience.

Secondary Details View

The detailed finance dashboard should be available, but it should never be the first screen.

If the owner clicks “View Details,” they can see:

Website Term	What It Means

New orders created	Orders likely caused by the SMS offer
Would-have-ordered-anyway risk	Customers who may have bought without the discount
Profit after offer	Profit remaining after food cost, discount, SMS, and fees
Campaign profit return	Whether the campaign made more profit than it cost
Test comparison group	Similar customers who did not receive the offer
Future repeat purchase value	Expected profit from customers returning again

This keeps the website clean while still proving that the financial model exists.

Website Pitch Language

The cleanest pitch is:

> Pielot looks at your POS data, finds your slowest revenue windows, and automatically recommends SMS offers that bring the right customers back without destroying your margins.

A shorter homepage version could be:

> Turn slow days into repeat visits with POS-powered SMS recommendations.

A restaurant owner should not feel like they are buying analytics software. They should feel like they are getting a clear revenue recommendation.

How to Explain the Financial Model Simply

The website can explain the model in plain language:

Pielot estimates how many new orders the offer should create, subtracts the cost of the discount, SMS, food, and payment fees, then shows the expected profit before you launch.

That is enough for the main page.

The backend can still calculate:

New Orders Created × Average Order Value = Incremental Revenue

Incremental Revenue
- Food/Product Cost
- Discount Cost
- SMS Cost
- Payment Fees
= Immediate Profit

Immediate Profit + Future Repeat Purchase Profit = 90-Day Profit Impact

But the owner-facing website should mostly show the final recommendation, not the math.

Personalized Offer Logic

Pielot should not treat the restaurant's customer base as one giant group.

It should tailor SMS offers based on individual customer behavior:

- What they usually buy
- How much they usually spend
- Which days they usually visit
- Whether they order alone or with others
- Whether they respond to offers
- Whether a larger minimum spend can increase their basket size

Individual Offer Example

Customer Type: Weekday slice buyer
Usual spend: $18
Suggested offer: $3 off $20+ on Tuesday afternoon
Goal: Move the customer into a slow window and increase basket size

Group Offer Example

Detected pattern: Customers often order together
Suggested offer: $8 off $45+ when redeemed as a group
Goal: Bring multiple customers into a slow window and increase total order value

The website should explain this carefully. Pielot should feel personalized, not invasive.

Use language like:

Pielot detects customer patterns and recommends smarter offers.

Avoid language like:

Pielot tracks exactly when specific people come in together.

Clean Website Pitch

Pielot is built for restaurant owners who do not have time to run complicated marketing experiments.

Instead of asking the owner to guess when to discount, Pielot reads the POS data, finds weak revenue windows, and recommends an SMS campaign with a clear profit estimate.

The owner sees one recommendation:

This day is slow.
This customer group is likely to return.
This offer should increase orders.
This is the expected profit.
Launch, edit, or reject.

That is the entire product experience.

The backend can handle the advanced financial modeling. The website should make the value feel obvious:

> Pielot turns POS data into margin-safe SMS campaigns that bring customers back on slow days.

How the Product Shows Expertise

The website should prove that the team understands restaurants by showing practical guardrails, not by using complicated finance language.

Pielot should feel credible because it protects the owner from bad campaigns.

It should show:

- Profit before launch
- Minimum spend thresholds
- Offer frequency limits
- SMS opt-in protection
- Test comparison groups
- Margin-safe recommendations
- Clear campaign results after launch

This tells the customer that Pielot is not just trying to send more texts. It is trying to help the restaurant make better revenue decisions.

Why Pleasure Pizza Is a Good Example

Pleasure Pizza is a strong example because it has the kind of local restaurant profile where repeat customers matter.

Publicly available information shows that Pleasure Pizza has a long-standing Santa Cruz presence, multiple customer-facing locations, dine-in, to-go, and delivery options, and online ordering connected to Toast for at least the Downtown location.

That matters because Pielot is designed for businesses that already have POS activity, repeat customers, and predictable demand swings.

Pleasure Pizza likely has natural revenue windows such as:

- Stronger weekend demand
- Beach and tourist traffic
- Lunch and dinner rushes
- Slower weekday afternoons
- Location-specific differences between Downtown, East Side, and Pleasure Point traffic

Pielot would use actual POS data to confirm these patterns instead of guessing.

The Problem

Restaurants do not struggle because every day is slow.

They struggle because demand is uneven.

Some windows are busy. Other windows are underused. Staff, kitchen capacity, and inventory are already available, but customer traffic is weaker than it could be.

Most restaurants respond with broad discounts.

That creates problems:

- Discounts may go to customers who would have ordered anyway
- Margins can shrink
- Customers may learn to wait for deals
- The owner may not know whether the campaign actually worked

Pielot solves this by making one clear recommendation at a time.

The Owner-Facing UI

The restaurant owner should not see a spreadsheet first.

The first screen should show five things:

1. Slow day detected
2. Recommended customer segment
3. Suggested offer
4. Expected profit
5. Launch / Edit / Reject

The finance model can live in the backend. The owner-facing experience should feel like an AI assistant making one clear recommendation.

Example Demo Screen

Tuesday Slow-Day Campaign

Slow window: Tuesday 2–6 PM
Target: At-risk weekday customers
Offer: $5 off $30+
Texts sent: 4,000
Expected redemptions: 320
Estimated new orders created: 192
Incremental revenue: $5,376

90-day profit impact: $3,169
Recommendation: Launch with test group

[Launch Campaign] [Edit Offer] [Reject]

This is the product.

Not a complicated dashboard.

A simple recommendation that the owner can understand in seconds.

Restaurant-Friendly Language

Pielot should avoid finance-heavy wording on the main screen.

Technical Term	Restaurant-Friendly Term
Incrementality rate	New orders created
Cannibalization risk	Would-have-ordered-anyway risk
Gross margin after discount	Profit after offer
Promotion ROI	Campaign profit return
Control group	Test comparison group

The detailed terms can still exist in the finance view, but the owner should see plain language.

How Pielot Works

Pielot connects to the POS system and looks for low-revenue patterns.

It uses data such as:

- Order date and time
- Ticket size
- Customer visit frequency
- Prior purchases

- Promo redemptions
- Discounts
- Refunds or voids
- Location
- SMS opt-in status

Then it recommends a campaign.

The workflow is:

POS Data
↓
Find slow revenue windows
↓
Pick the best customer segment
↓
Suggest a margin-safe SMS offer
↓
Estimate profit impact
↓
Owner approves, edits, or rejects
↓
Track actual results

What the Backend Calculates

The owner does not need to see every formula, but the backend needs to calculate them.

The most important financial logic is:

New Orders Created × Average Order Value = Incremental Revenue

Then:

Incremental Revenue
- Food/Product Cost
- Discount Cost
- SMS Cost
- Payment Fees
= Immediate Profit

Then Pielot adds future retention value:

Immediate Profit + Future Repeat Purchase Profit = 90-Day Profit Impact

The dashboard should show the result, not the entire spreadsheet.

Demo Walkthrough Script

Here is how I would explain the product in a presentation:

“Let’s say Pleasure Pizza usually has slower traffic on Tuesday afternoons from 2 to 6 PM. Pielot detects that slow window by looking at POS sales history.

Instead of telling the owner to discount everything, Pielot recommends a specific campaign.

For this example, the target is at-risk weekday customers — people who have ordered before during the week but have not come back recently.

The suggested offer is $5 off orders over $30.

Pielot estimates that if we send 4,000 texts, about 320 customers will redeem the offer. But the important number is not redemptions. The important number is new orders created.

Out of those redemptions, Pielot estimates 192 are new orders that likely would not have happened without the SMS.

That creates $5,376 in incremental revenue and an estimated 90-day profit impact of $3,169.

So the recommendation is simple: launch with a test comparison group.”

Why SMS Only

SMS keeps the product simple for local restaurants.

Restaurants do not need a complicated ad system. They need a fast way to reach customers when demand is weak.

SMS is useful because:

- It is immediate
- It is easy to understand

- It works for time-sensitive food offers
- It does not require customers to download an app
- It can connect directly to POS customer profiles and opt-in records

Pielot should feel like:

Slow day coming. Here is who to text. Here is the offer. Here is the expected profit. Launch?

Personalized and Group-Based Discounts

Pielot should not treat every customer the same.

Instead of sending one generic offer to the entire customer base, Pielot can tailor each discount based on the individual customer's POS behavior.

For example, if one customer usually visits on weekends and another customer usually visits on weekdays, they should not receive the same offer at the same time. Pielot should recommend the offer most likely to change each customer's behavior.

The model can look at:

- How much the customer usually spends
- Which products or categories they buy
- Which days and times they usually visit
- How often they return
- Whether they usually respond to discounts
- Whether they tend to order alone or with others
- Whether a larger offer could encourage a larger basket size

The goal is to incentivize additional spending, not just give away margin.

Individual Discount Example

Customer A
Usually visits: Friday evenings
Average spend: $18
Favorite item: Slices and drink
Suggested offer: $3 off $20+ on Tuesday afternoon
Goal: Move one visit into a slow window and increase basket size

Customer B
Usually visits: Sunday with family
Average spend: $42

Favorite item: Whole specialty pizza
Suggested offer: $5 off $35+ on Wednesday dinner
Goal: Bring back a higher-value customer during a weaker weekday

Friend and Group Discount Example

If the POS database frequently observes a few customers ordering together, Pielot can create group-based offers.

For example, if Sriram and Animesh often come in together or are part of the same ordering pattern, Pielot can recommend a group offer that encourages both people to redeem together.

Detected Pattern
Two customers often visit together on weekends.

Slow Window
Tuesday 2–6 PM

Suggested Group Offer
$8 off $45+ when both customers redeem together

Goal
Move a group visit into a slow window and increase total order value.

This makes the discount more strategic. The restaurant is not just rewarding one person. It is encouraging a larger group purchase during a low-demand window.

Why Group Discounts Matter

Group offers can increase:

- Total order size
- Visit frequency
- Customer retention
- Social purchasing behavior
- Low-demand day traffic

But Pielot should still protect margin. A group discount should only launch when the expected order size is high enough to justify the offer.

Why This Is Different

Most tools help restaurants send campaigns.

Pielot helps restaurants decide whether a campaign is worth sending and who should receive which offer.

A normal marketing tool says:

Send 10% off to your customer list.

Pielot says:

Tuesday 2–6 PM is projected to be slow.
At-risk weekday customers are the best target.
$5 off $30+ should create new orders while protecting profit.
Launch with a test comparison group.

That is the difference.

Pielot is not selling text messages.

Pielot is selling better slow-day revenue decisions.

What the Details View Shows

The full finance dashboard should be secondary.

The owner can click “View Details” to see:

- Texts sent
- Expected redemptions
- New orders created
- Would-have-ordered-anyway risk
- Incremental revenue
- Discount cost
- SMS cost
- Payment fees
- Profit after offer
- 90-day profit impact
- Campaign profit return
- Test comparison group results

This keeps the main UI simple while still giving financial transparency when needed.

Guardrails

Pielot should block or warn against bad campaigns.

A campaign should be flagged if:

- Profit after offer is too low
- Would-have-ordered-anyway risk is too high
- The customer segment has been texted too often
- The offer is too aggressive
- The SMS list has high unsubscribe risk
- The POS data is incomplete
- There is no test comparison group

This protects the restaurant from accidentally training customers to wait for discounts.

Closing

Pielot gives small restaurants a smarter way to use the data they already have.

For a business like Pleasure Pizza, the value is not just sending an SMS.

The value is knowing when demand is weak, which customers are worth contacting, what offer protects margin, and whether the campaign creates real profit.

The final pitch is:

> Pielot helps restaurants use POS data and SMS to turn slow days into profitable repeat-customer visits.

Not more random discounts.

Better timing. Better targeting. Better retention. Better slow-day revenue.

Tab 7

final

Pielot Production PRD Correction

Real Chat-Based AI Revenue Agent for Local Restaurants

1. Correct Product Direction

Pielot is a real production application, not a simulated demo.

It should still feel like a custom AI chat where the owner talks naturally to Pielot, but the actions behind the chat must be real:

- Real customer/order data upload
- Real data validation
- Real customer segmentation
- Real revenue opportunity detection
- Real SMS campaign generation
- Real compliance checks
- Real owner approval
- Real campaign creation
- Real SMS sending or scheduling
- Real campaign status tracking
- Real campaign page with live metrics

The chat should be the main interface, but it should operate real backend workflows.

---

2. Corrected Product Summary

Pielot is a real AI SMS revenue agent for local restaurants. Owners talk to Pielot through a chat-first interface and ask for ways to increase revenue, recover slow days, bring back old customers, or create campaigns.

Pielot uses connected or uploaded POS/customer data to find revenue opportunities, generate personalized SMS campaigns, check margin and compliance guardrails, and let the owner approve campaigns before they are sent.

Once approved, Pielot actually schedules or sends the campaign through an SMS provider, creates a real campaign page, and tracks delivery, clicks, opt-outs, redemptions, attributed orders, revenue, and profit impact.

---

3. What "Thinking" Means in the Real Product

The app should still show thinking/progress states, but they should not be fake animations disconnected from backend work.

Each visual progress state should map to a real backend job.

Example: User asks

"Find me a way to make more money this week."

Pielot shows progress

- Reading uploaded POS/customer data
- Validating opt-in status
- Segmenting customers
- Detecting low-demand windows
- Estimating revenue opportunity
- Checking margin guardrails
- Preparing campaign recommendation

What is actually happening

- Backend fetches customer/order data
- Data validation service checks required fields
- Segmentation service assigns customers to groups
- Revenue scan service identifies slow windows
- Campaign recommendation agent generates a structured recommendation
- Compliance guardrail service filters unsafe recipients
- Frontend displays the final insight card

The UI should make the work visible, but the work should actually happen.

---

4. Updated Technical Stack

Frontend

- Vite
- React

- TypeScript
- Tailwind CSS
- shadcn/ui-style components
- Framer Motion for animations
- React Router
- Zustand or React Context for state
- Recharts for analytics

Intelligence Layer

- MiniMax token plan for LLM intelligence
- Agents SDK for orchestration
- Structured JSON outputs for cards/campaigns
- Tool calls for backend actions
- Guardrails for compliance and approval

OpenAI's Agents SDK is built for applications that own orchestration, tool execution, approvals, and state; that fits Pielot because the product needs agents that can plan, call tools, use guardrails, and coordinate workflows before campaigns are approved or sent.

Backend

Recommended:

- Node.js / Express or FastAPI
- PostgreSQL or Supabase Postgres
- Redis or queue system for campaign jobs
- Background workers for SMS sending and attribution
- File storage for CSV uploads
- Webhooks for SMS events

SMS Provider

Production MVP should use one real SMS provider:

- Twilio, preferred for MVP
- Telnyx, optional alternative

Data Sources

MVP:

- CSV upload
- Google Sheets import
- Manual customer import

- Demo seed data only for testing, not core product

Future:

- Toast
- Square
- Clover
- DoorDash exports
- Uber Eats exports
- Shopify
- Stripe

---

5. Real MVP Scope

Must Be Real

The MVP must support:

1. Restaurant profile setup
2. CSV customer/order upload
3. Data validation
4. SMS opt-in filtering
5. Customer segmentation
6. AI revenue scan
7. AI-generated campaign recommendation
8. SMS copy generation
9. Margin/profit estimate
10. Compliance checklist
11. Owner approval before sending
12. Real campaign creation
13. Real SMS scheduling/sending
14. Real campaign status page
15. Real delivery/failed/opt-out tracking
16. Real zero-state before messages send
17. Real metrics update after events arrive

Can Be Simplified

The MVP can simplify:

- POS integrations can start with CSV

- Attribution can start with promo codes and links
- Profit estimates can use owner-provided margin assumptions
- Customer segmentation can use rules + AI explanation
- Campaign performance can start with delivery, clicks, opt-outs, and manually/imported attributed orders

Should Not Be Built Yet

- Full Toast/Square/Clover integrations
- Fully autonomous sending without approval
- Complex ML forecasting
- Multi-location enterprise analytics
- Full CRM replacement
- Billing
- Advanced A/B testing
- Inventory-aware campaigns
- Weather/event automation

---

6. Core Real Product Flow

Flow: Chat to Real Campaign

1. Owner logs in.
2. Owner creates restaurant profile.
3. Owner uploads customer/order CSV.
4. Pielot validates data.
5. Owner confirms SMS consent source.
6. Owner opens chat.
7. Owner asks: “Find me a way to make more money this week.”
8. Pielot runs a real revenue scan.
9. Pielot shows visual progress tied to backend jobs.
10. Pielot returns a real insight card.
11. Owner clicks “Create Campaign.”
12. Pielot creates a real campaign draft object.
13. Owner reviews SMS, audience, offer, and projected impact.
14. Owner clicks “Approve & Schedule.”
15. Pielot schedules SMS through provider.
16. Campaign page is created.
17. Initial metrics show zero until send begins.
18. SMS provider webhooks update delivery, failure, and opt-out data.

19. Campaign page updates live.
20. Owner sees actual campaign results.

---

7. Main Chat Interface

The chat remains the main interface.

Main Screen Layout

Left Sidebar

- Pielot logo
- New chat
- Campaigns
- Customers
- Segments
- Compliance
- Settings
- Restaurant status
- Data source status
- SMS provider status

Main Chat Area

Header:

Pielot

Subtitle:

AI revenue agent for your restaurant

Status pills:

- POS data connected
- SMS provider connected
- Consent verified
- Campaign approval required

Suggested Prompts

- Find me a revenue opportunity

- Create a campaign for lapsed customers
- What should I send this week?
- Why are Tuesdays slow?
- Make this offer less discount-heavy
- Show me campaign performance

Chat Input

Placeholder:

Ask Pielot to find revenue, create campaigns, or explain your customers...

---

8. Real Thinking/Progress System

Every thinking card should be backed by real job status.

Component: AgentProgressCard

Required fields

- workflow_id
- workflow_type
- current_step
- steps
- status
- progress_percent
- started_at
- completed_at
- error_message
- output_card_id

Step states

- pending
- running
- complete
- failed
- needs_approval

Example: Revenue Scan Progress

Title:

Pielot is scanning for revenue opportunities

Steps:

1. Loading customer/order data
2. Checking opt-in eligibility
3. Segmenting customers
4. Finding slow revenue windows
5. Matching offers to segments
6. Estimating revenue and margin
7. Running compliance guardrails
8. Preparing recommendation

Each step should update based on backend completion, not only frontend timers.

---

9. Backend Workflows

Workflow 1: Data Validation

Triggered when CSV is uploaded.

Inputs

- restaurant_id
- uploaded_file_id

Actions

- Parse file
- Validate required columns
- Normalize phone numbers
- Check SMS opt-in status
- Remove duplicates
- Detect missing values
- Store customers
- Store orders
- Generate data quality score

Output

- customers_found
- opted_in_customers
- opted_out_customers
- invalid_phone_numbers
- missing_consent_count
- data_quality_score
- validation_warnings

Workflow 2: Revenue Scan

Triggered from chat or dashboard.

Inputs

- restaurant_id
- date_range
- optional user goal

Actions

- Load recent orders
- Calculate weekday/hour revenue patterns
- Detect weak revenue windows
- Segment customers
- Identify campaign opportunities
- Rank opportunities
- Estimate revenue/profit potential
- Return top recommendation

Output

- opportunity_id
- slow_window
- segment
- segment_size
- recommended_offer
- estimated_revenue
- estimated_profit
- confidence
- reasoning

Workflow 3: Campaign Draft Creation

Triggered when user clicks “Create Campaign.”

Inputs

- opportunity_id
- restaurant_id
- selected_segment
- recommended_offer

Actions

- Create campaign draft
- Generate SMS copy
- Generate promo code
- Estimate cost
- Estimate profit
- Run compliance checks
- Save campaign draft

Output

- campaign_id
- status: draft
- audience_count
- sms_body
- offer
- send_time
- promo_code
- projected_metrics
- compliance_status

Workflow 4: Campaign Approval

Triggered when user clicks “Approve & Send” or “Schedule.”

Inputs

- campaign_id
- owner_id

- approval_confirmation

Actions

- Verify owner permissions
- Re-run compliance checks
- Lock audience
- Schedule SMS job
- Create campaign recipients
- Mark campaign as scheduled

Output

- campaign_id
- status: scheduled
- scheduled_at
- recipient_count

Workflow 5: SMS Sending

Triggered by scheduled campaign worker.

Inputs

- campaign_id

Actions

- Load campaign recipients
- Send SMS through provider
- Store provider message IDs
- Update recipient delivery status
- Mark campaign as sending/sent

Output

- sent_count
- failed_count
- provider_batch_id

Workflow 6: SMS Webhooks

Triggered by SMS provider events.

Inputs

- provider message event

Actions

- Match provider message ID to campaign recipient
- Update delivery status
- Track failed messages
- Process STOP/opt-out replies
- Update opt-out table
- Exclude opted-out customers from future campaigns

Output

- updated recipient status
- campaign metrics updated

Workflow 7: Campaign Attribution

MVP attribution can use:

- Unique promo code
- Unique tracking link
- Manual POS import after campaign
- Time-window matching

Inputs

- campaign_id
- promo redemptions
- order import
- click events

Actions

- Match orders to campaign
- Calculate attributed revenue

- Estimate profit impact
- Update campaign metrics

Output

- orders_attributed
- revenue_generated
- estimated_profit_impact
- redemption_rate

---

10. Custom Chat Cards

Card 1: Revenue Insight Card

Shown after a real revenue scan.

Fields

- opportunity_id
- slow_window
- demand_gap
- baseline_revenue
- predicted_revenue
- segment_name
- segment_size
- recommended_offer
- estimated_revenue
- estimated_profit
- confidence
- reason

Buttons

- Create Campaign
- Explain Insight
- View Details
- Ignore

Card 2: Campaign Draft Card

Shown after a real campaign draft is created.

Fields

- campaign_id
- campaign_name
- status: Draft
- target_audience
- audience_size
- offer
- promo_code
- send_time
- sms_preview
- projected_redemptions
- projected_orders
- projected_revenue
- projected_profit
- confidence
- compliance_checklist

Buttons

- Approve & Send
- Schedule
- Edit SMS
- Change Offer
- Change Audience

Card 3: Offer Comparison Card

Shown when user asks “Why this offer?”

Fields

- recommended_offer
- rejected_offers
- margin_risk
- projected_conversion
- reason

Buttons

- Keep Recommended Offer
- Try Another Offer
- Rewrite SMS

Card 4: Live Campaign Card

Shown after campaign approval/scheduling.

Fields

- campaign_id
- campaign_name
- status
- audience_size
- send_time
- promo_code
- delivery_status
- texts_sent
- delivered
- failed
- redemptions
- orders_attributed
- revenue_generated
- profit_impact
- unsubscribes

Buttons

- Open Campaign Page
- Pause Campaign
- Duplicate
- View Setup

---

11. Campaign Page

URL

`/campaigns/:campaignId`

Purpose

Show the real campaign object and performance.

Sections

Header

- Campaign name
- Status: Draft / Scheduled / Sending / Sent / Completed / Paused
- Campaign ID
- Created by
- Created date
- Approved by
- Scheduled send time

Campaign Setup

- Audience
- Audience size
- Segment logic
- Offer
- Promo code
- SMS body
- Send time
- Valid window
- Control/test group, optional

Live Performance

Before sending, values show zero.

- Texts sent: 0
- Delivered: 0
- Failed: 0
- Clicks: 0
- Replies: 0
- Redemptions: 0
- Orders attributed: 0
- Revenue generated: $0
- Estimated profit impact: $0
- Unsubscribes: 0

After webhooks/imports arrive, metrics update.

Projection vs Actual

Projection:

- Expected redemptions
- Expected orders
- Expected revenue
- Expected profit

Actual:

- Actual redemptions
- Actual orders
- Actual revenue
- Actual profit

Timeline

- Draft created
- Compliance passed
- Owner approved
- Scheduled
- Sending started
- Sending completed
- First delivery received
- First redemption received
- Results updated

Ask Pielot About This Campaign

Mini chat scoped to the campaign.

Example prompts:

- Why did we choose this audience?
- What happens if redemptions are low?
- Can we make this campaign safer?
- What should we test next?

---

12. Data Model

restaurants

- id
- owner_id
- name
- cuisine_type
- location
- timezone
- average_order_value
- brand_tone
- order_link
- sms_sender_name
- margin_assumption
- created_at
- updated_at

customers

- id
- restaurant_id
- first_name
- phone_number
- phone_hash
- sms_opt_in_status
- opt_out_status
- last_order_date
- order_count
- total_spend
- average_order_value
- favorite_items
- preferred_order_day
- preferred_order_time
- coupon_usage_score
- created_at
- updated_at

orders

- id
- restaurant_id
- customer_id
- order_date
- order_total

- discount_amount
- items
- channel
- promo_code
- created_at

segments

- id
- restaurant_id
- name
- description
- criteria_json
- customer_count
- created_at

campaigns

- id
- restaurant_id
- segment_id
- name
- status
- offer
- promo_code
- sms_body
- scheduled_at
- approved_by
- approved_at
- projected_revenue
- projected_profit
- actual_revenue
- actual_profit
- created_at
- updated_at

campaign_recipients

- id
- campaign_id
- customer_id
- phone_number

- provider_message_id
- delivery_status
- clicked
- replied
- ordered
- revenue_attributed
- opted_out_after
- sent_at
- delivered_at

opt_outs

- id
- restaurant_id
- phone_number
- keyword
- campaign_id
- opted_out_at

agent_runs

- id
- restaurant_id
- user_id
- workflow_type
- status
- input_json
- output_json
- started_at
- completed_at

agent_steps

- id
- agent_run_id
- step_name
- status
- started_at
- completed_at
- output_summary

---

13. Compliance Requirements

Pielot must be compliance-first.

Required SMS Guardrails

- Only send to opted-in customers
- Never send to opted-out customers
- Include opt-out language
- Process STOP replies automatically
- Enforce quiet hours
- Enforce frequency caps
- Require human approval before sending
- Keep approval logs
- Keep campaign audit logs
- Prevent sensitive targeting
- Prevent deceptive urgency
- Let owner pause campaigns

Default Frequency Caps

- Max 2 marketing texts per customer per week
- Max 6 marketing texts per customer per month
- No marketing texts before 9 AM local time
- No marketing texts after 8 PM local time
- No repeated campaign to same segment within 72 hours

---

14. Intelligence Requirements

The MiniMax/agent layer must produce structured outputs, not just prose.

Required structured output for campaign recommendation

{
  "campaign_name": "Tuesday Comeback Offer",
  "segment_name": "At-risk weekday customers",
  "customer_count": 1248,
  "segment_reason": "Customers who previously ordered on weekdays but have not returned in 21–45 days.",

```
  "offer": "$5 off orders above $30",
  "sms_body": "Pleasure Pizza: We miss you. Get $5 off your order of $30+ today from 2–6 PM. Order here: [link]. Reply STOP to opt out.",
  "send_time": "Tuesday 1:15 PM",
  "projected_orders": 60,
  "projected_revenue": 1680,
  "projected_profit": 730,
  "risk_level": "medium-low",
  "compliance_status": "ready_for_approval",
  "required_owner_action": "approve_before_send"
}
```

AI must not

- Hallucinate customer data
- Send without approval
- Target opted-out customers
- Generate messages without opt-out language
- Recommend offers that violate margin rules
- Use sensitive attributes
- Claim actual performance before data exists

---

15. Real Visual Progress Requirements

Visual progress should still exist, but the frontend must subscribe to real workflow state.

Implementation

When a workflow starts:

1. Backend creates `agent_run`.
2. Backend creates `agent_steps`.
3. Frontend subscribes using polling, SSE, or websockets.
4. Each step updates as backend work completes.
5. When final output is ready, frontend renders the corresponding card.

Example

User asks:

“Find me a way to make more money this week.”

Backend creates:

- agent_run: revenue_scan
- step 1: load_data
- step 2: validate_consent
- step 3: segment_customers
- step 4: detect_opportunity
- step 5: recommend_offer
- step 6: compliance_check
- step 7: generate_card

Frontend shows these as live thinking steps.

---

16. Acceptance Criteria

Pielot production MVP is complete when:

1. A restaurant can create a profile.
2. A restaurant can upload customer/order CSV data.
3. The system validates customer data and consent status.
4. The chat interface can trigger a real revenue scan.
5. The system can generate at least one real revenue opportunity from uploaded data.
6. The system can create a real campaign draft.
7. The system can generate SMS copy with opt-out language.
8. The system can run compliance checks.
9. The owner must approve before sending.
10. The system can schedule or send SMS through a real provider.
11. The system can process delivery and opt-out webhooks.
12. The system can create a campaign page.
13. The campaign page shows zero before sending.
14. The campaign page updates after real events arrive.
15. The agent activity/progress UI reflects backend workflow state.

---

17. Corrected Build Priority

Phase 1: Real MVP

Build:

- Vite React app
- Auth
- Restaurant profile
- CSV upload
- Data validation
- Customer/order database
- Chat UI
- Agent progress UI
- Revenue scan workflow
- Campaign draft workflow
- Approval workflow
- SMS provider integration
- Campaign page
- Webhook updates
- Compliance basics

Phase 2: Better Intelligence

Add:

- Better segmentation
- Offer comparison
- Margin modeling
- Promo code attribution
- Campaign recommendations
- Campaign learning

Phase 3: Integrations

Add:

- Google Sheets
- Toast
- Square
- Clover
- DoorDash/Uber Eats exports
- Multi-location support

Final Correct Product Principle

Pielot should feel magical, but it should not be fake.

The chat interface should make real backend workflows feel simple:

Ask Pielot → real data scan → real insight → real campaign draft → real approval → real SMS send/schedule → real campaign page → real metrics.

The “thinking” UI is not pretending. It is a visual layer over actual agent runs and backend jobs.